Great Financial Crisis and Dollar Dominance

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September 2026

Abstract

The majority of international trade transactions are invoiced in US dollars, with important implications for the global macroeconomy. This paper studies how temporary shocks to US dollar financing availability can generate persistent shifts in firms' invoicing decisions and their aggregate implications. Using Chilean export and import transactions from 2004 to 2019, we show that, following the Global Financial Crisis, the share of imports invoiced in US dollars fell by more than 8 percentage points, while the probability of invoicing in Chilean pesos increased permanently. We develop a model in which complementarities between trade invoicing and imported-input financing, combined with firm and currency-specific sunk costs, allow temporary changes in borrowing costs and UIP deviations to permanently alter invoicing choices. Consistent with the model, a temporary spike in US dollar financing costs induced firms to substitute toward the domestic currency, accounting for half of the aggregate shift. The effect is stronger for importers whose dollar financing needs are less naturally hedged by dollar-invoiced export revenues and weaker when competitors predominantly invoice in US dollars.

JEL Codes: F14, F31, F33, F41.

Keywords: Invoicing Currency, Dollar Dominance, Dominant Currency, Working Capital.

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1 Introduction

The global economy is dominated by the US Dollar, which serves as the dominant currency for invoicing international trade, pricing global assets, and holding reserves (Gopinath, 2015; Gopinath and Stein, 2021). Existing research explains this dominance through the supply of safe assets, complementarities between financing and invoicing decisions, and the role of the US dollar as a vehicle currency (Farhi and Maggiori, 2018; Devereux and Shi, 2013). Much less is known about the shocks that could weaken it. The evidence that a dominant currency can in fact be replaced comes almost entirely from the introduction of a policy: central bank swap lines that jump-started the renminbi (Bahaj and Reis, 2020; Chowdhry, 2024; Benguria and Novy, 2026), bilateral local-currency payment systems (Messer, 2022), the introduction of the euro (Benguria and Wagner, 2022), the political rupture of Brexit (Crowley et al., 2024; Garofalo et al., 2024), or the imposition of financial sanctions (Berthou, 2023; Chupilkin et al., 2025; Garofalo et al., 2025). Instead, we document a decline in the use of the dominant currency driven by a market event. Our source of variation is the Global Financial Crisis (GFC), which temporarily made borrowing in US dollars more expensive than borrowing in local currency, proxied by a reversal in deviations from uncovered interest parity (UIP). Firms moved their import invoicing out of the US dollar and into the domestic currency, and the shift outlasted the shock that produced it.

Using the universe of export and import transactions from Chile between 2004 and 2019, we show that, starting during the GFC, the share of imports invoiced in US dollars fell by nearly 10 percentage points, with a corresponding and permanent rise in the share of Chilean peso invoicing and no comparable movement on the export side. A decomposition of the observed change in aggregate US dollar invoicing attributes it to incumbent importers that reduce the share invoiced in US dollar within the origin-product pairs they already source, rather than to entry, exit, or reallocation across importers. The rise of the peso is driven by the same incumbents, both within continuing origin-product pairs and through the adoption of peso invoicing in new ones. The switch occurs at the firm-origin-product level and appears permanent: firms rarely revert once they switch, and both the probability of invoicing in pesos and the probability of switching from US dollar to peso invoicing remain permanently higher after the crisis.

We develop a theoretical framework to rationalize these empirical patterns. In our model, imported inputs serve as working capital that must be financed in advance (Bahaj and Reis, 2020). Firms face within-firm complementarities between the currency used for trade invoicing and the currency used to finance these imports. Differences in financing costs between the US dollar and local currency, proxied by UIP deviations, affect the invoicing choice by influencing the optimal borrowing currency. Invoicing in US dollars ties ex-post marginal costs to the realized exchange rate, which can lead to costly deviations from the optimal constant markup. As a result, firms opt for US dollar invoicing only when its financing advantage outweighs the risk of markup misalignment.

We also introduce currency- and firm-specific fixed costs of invoicing. These costs decline as more firms use the same currency, capturing strategic complementarities in invoicing (Crowley et al., 2020). Moreover, we assume that fixed costs include a sunk-cost component, such as the setup cost of managing transactions in a new currency: once incurred, the ongoing cost of using that currency remains permanently lower. Because of sunk costs, temporary shocks to financing in a particular currency, such as the US dollar during the GFC, can permanently change the relative appeal of different invoicing currencies, leading to lasting effects on firms' invoicing decisions, as observed in the data.

The model generates three key testable implications. First, when US dollar borrowing cost rises, firms become more likely to substitute the US dollar with the domestic currency. Second, conditional on rising borrowing costs in US dollar, firms whose financing needs in US dollar are less naturally hedged by dollar-invoiced revenues display a greater propensity to switch to domestic-currency invoicing. Third, this effect weakens when a larger share of competitors invoice in US dollar, reflecting stronger strategic complementarities in invoicing.

We test and verify the key predictions of our framework using import and export transactions from Chile. During normal times, borrowing in US dollar is typically cheaper than in domestic currency, but during the GFC the sign reversed, making peso borrowing more attractive. We then show that UIP deviations are a robust predictor of invoicing choices and account for roughly half of the aggregate rise in domestic currency invoicing. Also, consistently with our framework, importers whose dollar financing needs are less naturally hedged by dollar-invoiced export revenues respond more strongly when dollar financing becomes relatively more expensive, while firms more exposed to competitors invoicing in US dollar are less likely to switch, consistent with strategic complementarities in invoicing across importers.

Finally, we examine why the switch to the peso persists. According to the model, adopting a new invoicing currency requires the payment of a one-time sunk cost. Three pieces of evidence support this interpretation. First, importers that invoiced a larger share of their imports in pesos in the previous year are more likely to invoice in pesos today. Second, what matters is not only how intensively a firm has used the peso but whether it has ever used it: importers with any prior experience are substantially more likely to invoice in pesos again, and they respond more strongly when dollar funding becomes expensive, as expected of firms that have already paid the sunk cost of adoption. Third, the experience carries across the firm's markets: importers that have adopted the peso in one of their markets are more likely to adopt it in a market where they never have, consistent with a cost paid once at the firm level rather than in each trading relationship.

To conclude, we examine the macroeconomic implications of the decline in the aggregate share of dollar-denominated imports, particularly its effects on exchange rate pass-through and the sensitivity of the terms of trade and import inflation. Consistent with previous literature, invoicing plays a crucial role in determining the degree of exchange rate pass-through, which tends to be higher when transactions are invoiced in a dominant currency and lower when invoiced in a local currency (Gopinath et al., 2010; Barbiero, 2021). As a result, by

2019 the larger share of imports invoiced in local currency had reduced the sensitivity of import inflation to exchange rate fluctuations relative to 2007, while increasing the sensitivity of the terms of trade.

Related Literature Our paper relates to at least three strands of the literature. First, we contribute to the study of the forces that can alter the dominant currency equilibrium. Most models of currency dominance have in common the fact that they either define under which conditions a currency becomes dominant or they define under which conditions a currency remains dominant (Eichengreen et al., 2018; Chahrour and Valchev, 2022; Farhi and Maggiori, 2018; Gopinath and Stein, 2021).[1] Fewer papers instead focus on the shocks that can perturb the current dollar dominant equilibrium, and these have overwhelmingly relied on policy-driven variation. Bahaj and Reis (2020) study currency competition in the central banks' swap line market (see Bahaj and Reis, 2022, for a review); Chowdhry (2024) studies exporters' responses to the People's Bank of China internationalization reforms, and Benguria and Novy (2026) exploit the 2023 expansion of the China-Argentina swap line. Messer (2022) exploits the introduction of a bilateral local-currency payments system between Brazil and Argentina, and Benguria and Wagner (2022) the introduction of the euro while Mehl et al. (2023) study the shift to euro invoicing around EU enlargement. Crowley et al. (2024) and Garofalo et al. (2024) study the decline of sterling invoicing in UK exports after the Brexit referendum or Garofalo et al. (2025) show that the 2014 sanctions on Russia reduced the dominance of the US dollar in global banking; and, in theory, Chahrour and Valchev (2023) find that policies directly supporting the renminbi's use in trade could end dollar dominance.[2] Our paper complements this literature by showing that a dominant currency can lose ground without any such intervention: the variation we exploit is a market event, the relative cost of borrowing in the two currencies, moving for reasons unrelated to any policy introduction. Moreover, the policy-induced switches mentioned above moved real variables: Messer (2022) document an increase in export quantities under the Brazil–Argentina payment system while Chowdhry (2024) finds similar results in the context of French adopters of the renminbi, and similarly Argentine importers who switched currency under the swap line with the People's Bank of China temporarily imported more from China (Benguria and Novy, 2026). In our market-driven episode, by contrast, re-denomination leaves prices and quantities unchanged.

Second, our theoretical framework emphasizes the crucial role of trade finance. International trade transactions require external financing, which is typically heavily dollarized and sourced locally via domestic banks (Niepmann and Schmidt-Eisenlohr, 2017). The forces driving currency dominance in trade and finance are closely related. The availability of safe and liquid assets fosters dominance in the denomination of debt (Coppola et al., 2023), while

[1] Other recent contributions such as Gopinath et al. (2020) and Mukhin (2022) focus on understanding the consequences that the US dollar dominance has on international trade dynamics, and the global economy in general. Refer to Gourinchas et al. (2019) for a survey.

[2] Studying UK export prices around the same event, Corsetti et al. (2022) document large differences in pass-through across invoicing currencies but do not detect significant changes in the relative shares of invoicing currencies.

Eren and Malamud (2022) tie dollar debt dominance to the dollar being the riskiest major currency at typical debt maturities rather than the safest. Chahrour and Valchev (2022) highlight that dollar dominance arises from the need to post collateral to guarantee international trade transactions. Gopinath and Stein (2021) propose a feedback loop between invoicing and financing choices, in which exporters invoice in dollars because doing so lets them borrow in the cheaper currency, so that UIP deviations are central to the dominance of a currency. Xie (2024) instead considers invoicing goods for international trade in the presence of imperfect financial hedging of currency risk. While their contribution is theoretical, we empirically document how shocks to dollar financing, proxied by UIP deviations, can affect invoicing decisions.[3] Closer to us empirically, Kalemli-Ozcan and Varela (2021) and Salomao and Varela (2022) document how UIP deviations relate to capital flows and to firms' foreign currency borrowing, respectively. We complement these contributions by providing evidence that the scarcity of dollar-denominated trade finance, or, equivalently, large UIP deviations, during the GFC not only affected trade volumes (Bruno and Shin, 2019; Amiti and Weinstein, 2011), but also impacted invoicing decisions in international trade flows, with potential long-term consequences for dollar dominance.

Third, an extensive literature focuses on the firm-level determinants of invoicing choices and heterogeneous pass-through rates using micro-level data. Several papers have focused on the importance of variable markups and strategic complementarities in price-setting (Amiti et al., 2019, 2022; Berthou, 2023), international input intensity and operational hedging (Amiti et al., 2014; Nookhwun et al., 2025), and firm characteristics such as size and productivity (Berman et al., 2012).[4] Closest to our sunk-cost evidence, Crowley et al. (2020) show that a firm is more likely to invoice a new market in the dominant currency the longer it has used that currency in its existing markets, and rationalize this with a fixed cost of currency management. We document the mirror image of that mechanism: the same portability of currency experience across a firm's markets operates in favor of the currency that displaces the dominant one.[5] We show that, on one hand, financial considerations are a key determinant of invoicing choices. On the other hand, standard firm-level margins are crucial not only for understanding cross-sectional differences in invoicing patterns but also for explaining the adoption of different currencies.

Sections 2 and 3 describe the data and invoicing dynamics in Chile, respectively. Section 4 presents the theoretical framework. Section 5 provides evidence consistent with the key theoretical mechanisms. Section 6 assesses the aggregate implication.

[3]Our framework abstracts from financial hedging: firms could in principle offset the exchange rate risk of dollar invoicing by trading forwards (Lyonnet et al., 2022). Alfaro et al. (2021) also documents that the Chilean hedging market deepened considerably from the early 2000s. Section 5 shows that our results are unchanged when dollar funding shocks are measured by deviations from covered rather than UIP.

[4]See Burstein and Gopinath (2014), Itskhoki (2021), and Gopinath and Itskhoki (2022) for the most recent survey articles on this vast literature.

[5]Benguria and Novy (2026) reach a related conclusion in the Argentine episode, where renminbi use for imports from China spills over to imports from other origins, which they read as evidence that the fixed cost has a firm-currency component.

2 Data

Customs and Invoicing Data The primary data source is the universe of Chilean trade transactions from the Chilean Custom Agency (Aduanas). The agency provides transaction-level data for each firm, product, destination/origin, currency, and day, covering both export and import flows. The dataset includes the quantity exchanged (in kilograms and the product's unit of measure), the 8-digit HS classification of goods, and the transaction value (FOB) in US dollars. Crucially, it also records the invoicing currency, covering major currencies (US dollar, Euro, Chilean peso, UK pound, and Chinese yuan) as well as regional Latin American currencies. Additionally, an anonymous firm identifier allows for constructing a panel of Chilean importers and exporters. Import transaction data are available from 2007 to 2019, while export transaction data cover the period from 2009 to 2019. Since our main interest is the impact of the Global Financial Crisis on the use of the US dollar in international trade, we supplement the Chilean Customs Agency dataset with a similar dataset from Garcia-Marin et al. (2019).[6] This additional dataset contains the same information as the official customs records starting from 2004, allowing us to capture the pre-GFC period and rule out potential pre-trends. The dataset covers customs transactions from Chile's main trading partners. It accounts for 98% of total Chilean trade in value and quantity compared to official data, which give us confidence in the quality and the validity of the data.

UIP Data To proxy for domestic and foreign borrowing costs in constructing UIP deviations between the U.S. dollar and the Chilean peso, we use publicly available data from Chile's financial regulatory authority, the *Comisión para el Mercado Financiero* (CMF), which collects and reports aggregate data on the Chilean banking sector. They provide average deposit and lending rates at monthly frequency, distinguishing between domestic and foreign currency-denominated loans, as well as variations by maturity, loan size, and client type (commercial vs. non-commercial). We focus on commercial lending rates for short-term (one year) loans denominated in Chilean pesos and US dollars. Short-term loan rates are particularly relevant, as trade financing for international transactions heavily depends on this type of credit (Amiti and Weinstein, 2011; Ahn et al., 2011; Schmidt-Eisenlohr, 2013). The data cover the period around the GFC but are only available until 2014, when regulatory changes disrupted consistent comparisons between domestic and US dollar-denominated loans.

We complement the data on borrowing costs with data on the bilateral exchange rate between the U.S. dollar and the Chilean peso. We use the U.S. dollar-peso spot exchange rate obtained from Datastream to construct our measure of realized deviations from UIP. We also construct UIP deviation relying on monthly one-year-ahead exchange rate forecasts from Bloomberg. The dataset begins in June 2006 and includes forecasts of the U.S. dollar-peso exchange rate from approximately 160 financial institutions and market makers. We aggregate these forecasts by taking their cross-sectional average at the quarterly frequency.[7]

[6] We kindly thank Santiago Justel for sharing the dataset with us.
[7] Table 11 in Appendix A reports summary statistics for the exchange rate forecasts.

We use non-deliverable forward outright contracts for the U.S. dollar-peso from Bloomberg to construct measures of covered interest parity (CIP).

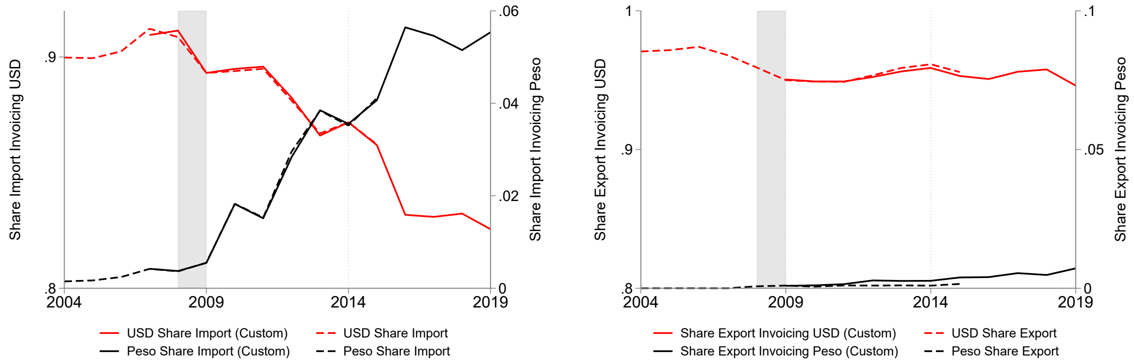
Lastly, we construct measures of realized UIP deviations relative to the US dollar for a panel of 113 countries. Interest rates and bilateral exchange rates are from the IMF International Financial Statistics, and US benchmark rates from FRED. We combine these series with country-level data on local currency invoicing in imports from Boz et al. (2025). The two sources overlap for 56 countries, which form the estimation sample. We use this information in a cross-country analysis in Appendix A.13.

3 Macro and Micro Dynamics in Invoicing

In this section, we use firm-level customs data to document three main empirical facts about invoicing dynamics in Chilean trade. First, we show that, following the GFC, the aggregate share of imports invoiced in U.S. dollars declined, while invoicing in Chilean peso increased, with no changes on the export side. Second, using a decomposition in the spirit of Melitz and Polanec (2015), we find that these shifts were primarily driven by incumbent importers, rather than by entry, exit, or reallocation across importers, which reduced their reliance on the U.S. dollar and increased invoicing in local currency. Third, we show that both the probability of invoicing imports in local currency and the probability of switching from U.S. dollar to local-currency invoicing permanently increased after the GFC.

Aggregate. Following the GFC, the share of imports invoiced in U.S. dollars declined by nearly 10 percentage points, while no change in invoicing patterns occurred on the export side. Figure 1 shows that, at the onset of the GFC, the share of imports denominated in U.S. dollars was approximately 90 percent. Over the following ten years, it declined to just above 80 percent. At the same time, the share of imports invoiced in Chilean peso increased from virtually 0 percent in 2004 to approximately 6 percent in 2019. Notably, we do not observe similar changes in the share of exports invoiced in different currencies as the share of exports denominated in the main currencies (U.S. dollar, Euro, and Chilean peso) remained stable over time.

Figure 1: Aggregate Invoicing Shares for Imports and Exports



Note: The Figure shows the time series of the aggregate share of imports denominated in US dollars and Chilean pesos for Chile between 2004-2019. The left vertical axis of Figure 1 reports the share of Chilean imports denominated in US dollar. The right axis shows the share of Chilean imports denominated in Chilean pesos. Data from 2007 (2009) for imports (exports) are from Chilean Customs Agency. Data before 2007 (2009) are from Garcia-Marin et al. (2019). The grey shaded area represents the NBER recession period.

Firm Level Decomposition. To understand the margins driving the dynamics of the aggregate invoicing shares reported in Figure 1, we perform a dynamic Olley-Pakes decomposition in the spirit of Melitz and Polanec (2015). The goal is to quantify the relative role played by: (i) firms' entry and exit and the reallocation of imports between and within firms; (ii) firms' sourcing strategies at the origin $\times$ product level; and (iii) invoicing currency adjustment margins.

Let $\mathrm{y}^k_{\eta,t}$ denote imports of firm η invoiced in currency k at time t, and $\mathrm{y}_{\eta,t} \equiv \sum_k \mathrm{y}^k_{\eta,t}$ is η's total imports. Thus, we can define $\lambda^k_{\eta,t}$ as the share of η's imports invoiced in currency k, $\lambda^k_{\eta,t} \equiv \frac{\mathrm{y}^k_{\eta,t}}{\sum_k \mathrm{y}^k_{\eta,t}} = \frac{\mathrm{y}^k_{\eta,t}}{\mathrm{y}_{\eta,t}}$. Let γ index the set of firms containing η, $\mathrm{y}_{\gamma,t} \equiv \sum_{\eta\in\gamma} \mathrm{y}_{\eta,t}$. Thus, we can define η's share of total imports at level γ as $s^\eta_{\gamma,t} \equiv \frac{\mathrm{y}_{\eta,t}}{\mathrm{y}_{\gamma,t}}$. For a given variable x, define $\overline{x_t} \equiv \frac{1}{2}(x_t + x_{t-1})$ and $\Delta x_t \equiv x_t - x_{t-1}$. Throughout this section, we use i to denote importers and m to denote origin-product pairs, with the product defined at the HS8 level.

Thus, the change in the share of total imports invoiced in currency k at time t, $\Delta\lambda^k_t$, can be decomposed as follow:

$$\Delta\lambda^k_t = \underbrace{\sum_{i\in I_t} \Delta s^i_t \overline{\lambda^k_{i,t}}}_{\text{Incumbent - Between}} + \underbrace{\sum_{i\in I_t} \Delta\lambda^k_{i,t}\overline{s^i_t}}_{\text{Incumbent - Within}} + \underbrace{\sum_{i\in I^+_t} \lambda^k_{i,t}s^i_t - \sum_{i\in I^-_t} \lambda^k_{i,t-1}s^i_{t-1}}_{\text{Net Entry}}, \tag{1}$$

$$\Delta\lambda^k_{i,t} = \underbrace{\sum_{m\in I_{it}} \Delta\lambda^k_{im,t}\overline{s^m_{i,t}}}_{\text{Within Origin-Product}} + \underbrace{\sum_{m\in I_{it}} \overline{\lambda^k_{im,t}}\Delta s^m_{i,t}}_{\text{Between Origin-Product}} + \underbrace{\sum_{m\in I^+_{it}} \lambda^k_{im,t}s^m_{i,t} - \sum_{m\in I^-_{it}} \lambda^k_{im,t-1}s^m_{i,t-1}}_{\text{Net Entry Origin-Product}} \quad \forall i\in I_t, \tag{2}$$

$$\Delta\lambda^k_{im,t} = \underbrace{\lambda^k_{im,t}\mathbf{1}_{\{A^k_{im,t}\}} - \lambda^k_{im,t-1}\mathbf{1}_{\{B^k_{im,t-1}\}}}_{\text{Currency Net Entry}} + \underbrace{\Delta\lambda^k_{im,t}\mathbf{1}_{\{C^k_{im,t}\}}}_{\text{Within-Currency}} \quad \forall i\in I_t, \forall m\in I_{i,t}, \tag{3}$$

where I_t is the set of continuing importers, while I^-_t and I^+_t denote the sets of exiting and entering importers; similarly, I_{it}, I^-_{it}, and I^+_{it} denote the sets of origin-product pairs m from which importer i imports in both t and $t-1$, only in $t-1$, and only in t, respectively.

Table 1: Olley-Pakes Decomposition of Imports Invoicing

	US dollar	Chilean peso	Euro	Others
Δ Aggregate Invoicing Share	-8.397	5.114	2.919	0.364
Net Entry	-1.794 (21.4%)	0.822 (16.1%)	0.808 (27.7%)	0.164 (45.1%)
Incumbents	-6.603 (78.6%)	4.292 (83.9%)	2.111 (72.3%)	0.200 (54.9%)
Incumbents	-6.603	4.292	2.111	0.200
Within Firm	-4.403 (66.7%)	5.151 (85.7%)	-0.780 (21.3%)	0.032 (16.1%)
Between Firm	-2.200 (33.3%)	-0.859 (14.3%)	2.891 (78.7%)	0.168 (83.9%)
Within Firm	-4.403	5.151	-0.780	0.032
Net Entry Origin-Product	1.761 (22.2%)	0.960 (18.6%)	-2.918 (57.7%)	0.196 (31.1%)
Within Origin-Product	-5.300 (66.9%)	3.663 (71.1%)	1.936 (38.3%)	-0.299 (47.4%)
Between Origin-Product	-0.864 (10.9%)	0.527 (10.2%)	0.201 (4.0%)	0.135 (21.4%)
Within Origin-Product	-5.300	3.663	1.936	-0.299
Currency Net Entry	-2.233 (42.1%)	2.107 (57.5%)	1.870 (96.6%)	-0.326 (92.3%)
Within Currency	-3.066 (57.9%)	1.556 (42.5%)	0.066 (3.4%)	0.027 (7.7%)

Notes: The Table reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin-product level for Chilean imports between 2007 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. We report the cumulative decomposition. Numbers are in percentage. Percentages in parentheses give each component's absolute value as a share of the sum of absolute values within its layer. Data are from the Chilean Customs Agency.

The first layer of the decomposition, Equation (1), allows us to assess whether the change in the aggregate share of imports invoiced in currency k is driven by net entry of importers or incumbent importers. For the latter, we disentangle the role played by changes in incumbent market shares from changes in their invoicing choices (within-firm dynamics versus between-firm reallocation). The second layer in Equation (2) assesses whether the observed change in incumbents' within margin is driven by shifts in sourcing across origin-product pairs. Finally, at the most granular level, Equation (3) shows that the share of imports invoiced in currency k in a given origin-product pair m can be decomposed into whether: (i) currency k is adopted in t but was not used in the preceding period $t-1$, indicating new adoption ($\mathbf{1}_{\{A^k_{im,t}\}}$); (ii) currency k is dropped in t after having been used in $t-1$ ($\mathbf{1}_{\{B^k_{im,t-1}\}}$); and (iii) there exists a change in the use of currency k when the importer continues using it across both periods ($\mathbf{1}_{\{C^k_{im,t}\}}$).

Table 1 reports the decomposition of the cumulative change in invoicing shares in Chilean imports between 2007 and 2019 across the four invoicing currencies: US dollar, Chilean peso, Euro, and the Other category grouping all remaining currencies. Close to 80 percent of the cumulative decline in the aggregate US dollar share is explained by incumbent firms ($-6.603/-8.397 \approx 0.79$), with the remainder due to net entry. Among incumbents, about two-thirds of the contribution ($-4.403/-6.603 \approx 0.67$) comes from the within-firm component, with the rest due to between-firm reallocation. Thus, the main driver is not that US dollar-invoicing importers are losing market share, but rather that they invoice less frequently in US dollars. Similarly, the increased use of the Chilean peso is almost entirely driven by incumbents that start invoicing more of their transactions into pesos ($5.151/6.010 \approx 0.86$).

Further zooming in at the origin-product level, we find that the bulk of the within-firm decline in US dollar invoicing arises within continuing origin-product pairs: importers keep sourcing the same origin-product pairs but invoice a smaller share in US dollars over time. By contrast, the rise in Chilean peso use is primarily driven both by the intensive margin – greater invoicing in pesos within continuing origin-product pairs ($3.663/5.151 \approx 0.71$) – and with a smaller contribution from the extensive margin – the adoption of peso invoicing in new origin-product pairs ($0.960/5.151 \approx 0.19$).

The bottom panel shows that both net currency switching and within-currency adjustments contribute sizably to the decline in the within origin-product component of US dollar invoicing, in a 42–58 split. That is, firms both drop the US dollar in favor of alternative currencies (Currency Net Entry) and, conditional on continuing to use the US dollar, reduce the share invoiced in it (Within-Currency). A symmetric interpretation applies for the Chilean peso. Importers increasingly invoice existing origin-product imports in Chilean pesos (Within-Currency), and are more likely to use the domestic currency when adopting a new currency (Currency Net Entry).

Lastly, Table 13 in Appendix A applies the benchmark decomposition to exports. The absence of significant aggregate movements reflects the fact that each of the underlying margins (net firm entry, within-firm adjustments, and between-firm reallocation) shows only modest changes. Thus, aggregate stability does not mask the presence of large but offsetting forces; rather, it reflects the fact that all underlying components changed only minimally, resulting in a muted overall changes in invoicing patterns.

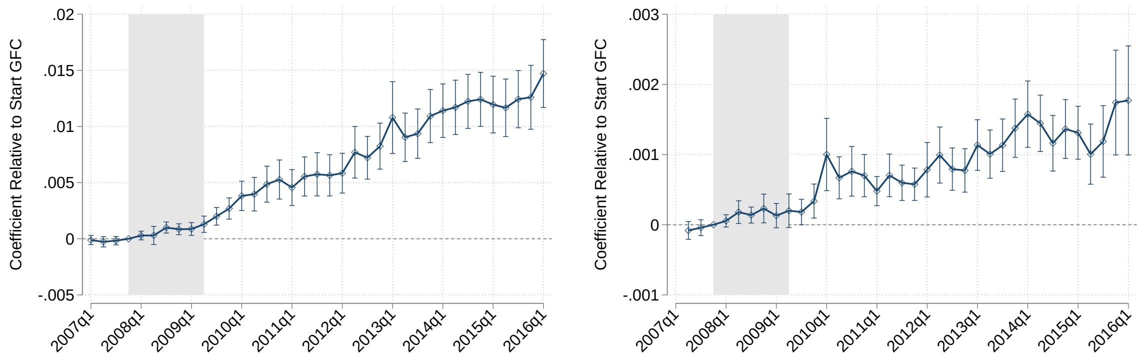
Overall, Table 1 delivers a clear message for interpreting the micro-level drivers of Figure 1. The decline in the aggregate US dollar invoicing share is mainly driven by incumbent importers, which reduce the share of their imports invoiced in US dollars at the origin-product level without significantly changing the quantities they import at that level. The opposite pattern emerges for invoicing in Chilean peso: incumbent importers invoice a substantially larger share of their imports in local currency, again without significant changes in imported quantities.

Probability of Invoicing and Switching after the GFC. We estimate whether the likelihood of invoicing in Chilean peso relative to the U.S. dollar has evolved following the GFC. To do so, we estimate the following specification:

$$y_{fopt} = \sum_{s=2007Q1}^{2016Q4} \beta_s \cdot \mathbf{1}[t=s] + \gamma_{fop} + \epsilon_{fopt}, \tag{4}$$

where the dependent variable is a dummy equal to one when a transaction at the $fopt$ level is invoiced in Chilean peso and zero when invoiced in U.S. dollars, i.e. $\mathbf{1}\{\text{Peso}_{fopt}\}$. As an alternative outcome, we consider a dummy equal to one if a firm invoices a transaction at the fop level in U.S. dollars at time $t-1$ and subsequently invoices the same transaction in Chilean peso at time t. The explanatory variables consist of a full set of quarter-year fixed effects spanning the period from 2007Q1 to 2016Q4. Our preferred specification includes

Figure 2: Effect of GFC on Probability of Invoicing in Chilean peso



Note: The Figure plots the estimated coefficients from Equation (4). Each point represents the effect of a given quarter on the probability of invoicing in Chilean peso, relative to the start of the GFC (2007Q4). The left panel reports results for the full sample, while the right panel restricts the sample to single-currency importers that switch from US dollar invoicing to Chilean peso invoicing. The gray dashed area corresponds to the NBER recession period (2007Q4-2009Q2). 95 percent confidence interval with standard errors clustered at the importer-origin level are reported. Data are from the Chilean Customs Agency.

fop fixed effects, allowing us to exploit within firm-HS8-origin variation to identify changes in the probability of invoicing in Chilean peso and in the likelihood of switching currency. This level of disaggregation is appropriate given that we observe meaningful variation at this level in the decomposition exercise.[8]

We find that both the probability of invoicing in and switching to Chilean peso increased permanently after the GFC. The probability of invoicing in Chilean peso relative to the US dollar increases steadily, reaching 2 percentage points above the pre-GFC level by the first quarter of 2016 (left panel of Figure 2). A similar upward trajectory is observed for the subset of switching firms, where the probability of switching from the US dollar to the Chilean peso is just below 0.2 percentage points higher at the beginning of 2016 than before the GFC (right panel of Figure 2). The effect is quantitatively large considering that, before the GFC, the unconditional probability of invoicing in Chilean peso is 0.7 percent, and the unconditional probability of switching is 0.04 percent.[9] Importantly, both panels show the absence of pre-trends before the GFC, indicating that the structural change in invoicing choices took place around the GFC.

Moreover, the switching to local currency invoicing is permanent. Figure 10 in Appendix A shows that, one year after the initial switch from US dollar invoicing to local currency invoicing, roughly 80% of the transactions at the importer-HS8-origin level continue to be invoiced in Chilean peso. This indicates that firms' invoicing choices are highly persistent over time and the structural change triggered by the GFC produced long-lasting effects. Figure 11

[8]Consistent with this choice, Table 9 in Appendix A shows that multi-currency invoicing within a firm-origin-HS8-year ($fopt$) is rare. Specifically, 96% of $fopt$ transactions are invoiced in a single currency, and a similar pattern holds when considering the share of import value. Invoicing in two currencies accounts for almost all of the remainder. The prevalence of single-currency invoicing at the $fopt$ level indicates that firms do not rely on combinations of currencies to implement their sourcing strategies. Table 10 in Appendix A shows that similar quantitative patterns hold on the export side.

[9]In line with these findings, Figure 9 in Appendix A show that the yearly transition probability from US dollar to Chilean peso invoicing more than tripled after the GFC, while transitions from the US dollar toward all other currencies declined.

in Appendix A confirms this pattern at annual frequency, where the data extend the pre-GFC window back to 2005.

Robustness: Invoicing Shares over Time Figures 7 and 8 in Appendix A plot the time-series contributions between 2007 and 2019 for each layer of the decomposition for imports invoiced in Chilean peso and US dollar, respectively. The figures show that the decline (increase) in the share of imports invoiced in US dollar (Chilean peso) is not concentrated in a particular year but instead evolves smoothly over time following the GFC. The conclusions drawn in the main section are robust, as they apply not only to the cumulative decomposition reported above, but also to the contribution of each layer at every point in time.

Robustness: Heterogeneity across Products and Origins Figure 12 in Appendix A shows that Chilean peso invoicing dynamics were similar across BEC categories (consumption goods, capital goods, and intermediate goods) and major trading partners (United States, Europe, China, Central-South America, Asia-Middle East, Africa, and Others).

We estimate Equation (4) separately for the top twelve origins and HS2 codes in the sample ranked by number importer-market observations. Figures 13 and 14 in Appendix A report the results. The average change in invoicing switching probability is not driven by a specific origin or product imported by Chilean firms, but a common feature across main trading partners and most relevant traded goods.

Moreover, Table 12 in Appendix A presents results from a dynamic Olley-Pakes decomposition with entry and exit at the origin level, abstracting from the product dimension. These results are qualitatively consistent with our benchmark decomposition.

4 A Model of Currency Choice

In this section, we present the theoretical framework used to interpret the empirical invoicing patterns observed among Chilean importers. Our model combines and extend two key mechanisms. First, we follow Bahaj and Reis (2020) in modeling within-firm complementarities between the currency used to invoice international trade transactions and the currency in which firms denominate their borrowing costs. Because of these complementarities, temporary shocks to US dollar financing availability (such as the GFC) alter the incentives to invoice in US dollar. Second, drawing on Crowley et al. (2020), we adopt a reduced-form specification of firm and currency-specific fixed costs of invoicing. We extend this fixed-cost structure by introducing sunk costs associated with adopting a new invoicing currency. The presence of sunk costs generates hysteresis in invoicing decisions following temporary shocks to financing costs, allowing the model to account for the persistence in invoicing patterns documented in Section 3.

4.1 Environment

A small open economy is populated by a continuum of importing firms indexed by $i \in [0,1]$. Each firm produces a differentiated good sold domestically and sets prices in domestic currency. Demand for firm i's output in the domestic market is given by $y_i^d = \left(\frac{p_i^d}{q^d}\right)^{-\theta}$, where p_i^d is the firm's price, q^d is a market-specific demand shifter, and $\theta > 1$ is the constant elasticity of demand.

Production combines domestic labor, l_i, and imported intermediate inputs, x_i, according to a constant-returns-to-scale Cobb–Douglas technology:

$$y_i = A_i l_i^{1-\alpha_i} x_i^{\alpha_i}, \tag{5}$$

where $\alpha_i \in (0,1)$ denotes the firm-specific share of imported inputs in production, and $A_i \equiv \left(\frac{1}{\alpha_i}\right)^{\alpha_i} \left(\frac{1}{1-\alpha_i}\right)^{1-\alpha_i}$ is a normalization constant. Imported inputs are assumed to be working capital and must be paid for prior to production, whereas labor costs are settled after revenues are realized.

Time is divided into two subperiods within each period: a morning and an evening. In the morning, firms choose prices, which are nominally rigid and remain fixed throughout the period, and finance the imported inputs required for production. Financing can be obtained either in domestic currency d or in a vehicle currency v. In the evening, firms purchase inputs, produce to meet demand at the preset price, collect revenues, and repay their loans.

A key feature of the environment is that the bilateral exchange rate between the domestic currency and the vehicle currency, denoted by s_v, is realized only after financing decisions are made. Consequently, fluctuations in the exchange rate affect the realized cost of imported inputs and can generate deviations between preset prices and ex post marginal costs.

Firms face the following production technology for imported inputs x_i:

$$x_i = \min_{\eta_i} \left(\frac{x_i^d}{\eta_i}, \frac{x_i^v}{1-\eta_i} \right). \tag{6}$$

By choosing η_i, the firm determines the relative shares of two imported inputs: x_i^d, denominated in domestic currency d, and x_i^v, denominated in the vehicle currency v. It follows from standard theory that firms optimally choose corner solutions for invoicing. In particular, imported inputs are entirely invoiced in either domestic or vehicle currency, so that $\eta_i^\star \in \{0,1\}$. In the corner solutions, if $\eta_i^\star = 0$, the firm invoices imported inputs exclusively in the vehicle currency, whereas if $\eta_i^\star = 1$, it invoices imported inputs entirely in domestic currency.

The firm must borrow to finance imported inputs and can choose between domestic and foreign currency borrowing. In principle, a firm could finance its imported inputs in a currency different from the one used to invoice international trade transactions. Following Bahaj and Reis (2020), however, we restrict attention to the case in which the financing currency coincides with the invoicing currency of imported inputs. This simplification is without loss of generality, as it arises optimally: firms prefer to align financing and invoicing currencies to minimize the exchange rate risk created by currency mismatches. Hence, in the model the currency chosen to invoice borrowing costs is equivalent to the currency used to invoice imported inputs.

The firm faces two financing options: (i) borrowing one unit of domestic currency and repay $(1 + i_d)$, or (ii) borrowing one unit of vehicle currency and repay $(1 + i_v)$. While interest rates are known at the time of borrowing, the exchange rate between the domestic and vehicle currencies, s_v, remains uncertain.

Given the cost structure described above, the *ex post* marginal cost of production for firm i, expressed in domestic currency, is:

$$C_i(s^v, \eta_i) = \frac{1}{A_i}\left[\frac{\eta_i \times \rho_d(1+i_d) + (1-\eta_i) \times \rho_v s_v(1+i_v)}{\alpha_i}\right]^{\alpha_i}\left[\frac{w}{1-\alpha_i}\right]^{1-\alpha_i}, \tag{7}$$

where w denotes the domestic wage rate, and ρ_d and ρ_v are the prices of domestic and imported intermediate inputs expressed in domestic and vehicle currency, respectively. All prices and borrowing costs are taken as exogenous and common across firms.

Lastly, we turn to the final key cost component: currency- and firm-specific fixed costs. Firms invoicing international transactions in currency j incur a fixed cost F_i^j. Following Crowley et al. (2020), we assume that this cost decreases with the number of firms using currency j. We extend their specification by introducing sunk costs in the adoption of new currencies, implying that a firm's fixed cost of using currency j depends on its invoicing choice in the previous period.

Formally, the reduced-form representation of firm i's fixed cost of invoicing in currency j at time t is:

$$F_{it}^j = f_{it}^j - \gamma^j\, \omega_{t-1}^j(N_{-it}), \tag{8}$$

where $\omega_{t-1}^j(N_{-it})$ denotes the share of importers using currency j in period $t-1$. The firm-specific component is given by

$$f_{it}^j = \begin{cases} \kappa_0^j & \text{if firm } i \text{ uses currency } j \text{ for the first time,} \\ \kappa_1^j & \text{if firm } i \text{ has previously used currency } j, \end{cases}$$

with $\kappa_1^j < \kappa_0^j$. The difference $\kappa_0^j - \kappa_1^j$ represents the sunk entry cost of invoicing in a new currency, capturing one-time managerial costs such as setting up foreign-currency bank accounts.[10]

The second term in Equation (8) captures externalities in adoption: the fixed cost decreases as more firms choose currency j, reflecting strategic complementarities in invoicing (Amiti et al., 2019; Crowley et al., 2020).[11] The strength of these complementarities is gov-

[10] We assume that the firm-specific component depends only on the previous period's currency choice. Richer specifications could allow a longer invoicing history to matter, leading to gradual cost accumulation. See Alessandria et al. (2021) for a review of sunk costs in the trade literature.

[11] In our framework, complementarities operate across firms. Empirical evidence in Section 3 shows that invoicing switching is not driven by specific origin-product pairs within firms. Nonetheless, the model can be

erned by γ^j, which satisfies the stability condition $0 < \gamma^j < f^j_{it} < 1$. The core idea of complementarities is that usage rises as overall utilization increases. One practical rationale is that firms may prefer to invoice in the same currency as their competitors to maintain similar exchange rate pass-through to import prices.

4.2 Working Capital Invoicing Choice

In this section, we solve for both the optimal invoicing decision for imported inputs and the optimal domestic pricing decision. Crucially, we obtain key predictions that we will later test directly in the data.

Optimal Invoicing Decision Firms are assumed to be risk-neutral and aim to maximize expected profits. They form expectations about the future realization of the exchange rate between the domestic currency and the vehicle currency, s_v. Fluctuations in exchange rate realizations affect the ex-post cost of borrowing in the vehicle currency, generating deviations from the optimal markup over marginal costs and reducing profits.

Given the isoelastic demand function and the imported-input technology in Equation (6), and normalizing the demand shifter q_d to one without loss of generality, firm i maximizes ex-ante profits:

$$\max_{\eta_i, p_i^d} \mathbb{E}\left[\pi_i^j\right] = p_i^d \left(p_i^d\right)^{-\theta} - \mathbb{E}\left[C_i(s_v, \eta_i)\right] \left(p_i^d\right)^{-\theta} - \eta_i F_i^d - (1-\eta_i) F_i^v. \tag{9}$$

For a given η_i, the optimal domestic price is $p_i^{d\star} = \frac{\theta}{\theta-1}\mathbb{E}[C_i(s_v, \eta_i)]$. Given $p_i^{d\star}$, firm i chooses to invoice imported inputs in the vehicle currency ($\eta_i^\star = 0$) if the following condition holds:

$$\eta_i^\star = 0 \qquad \text{iff} \qquad \mathbb{E}\left[(\rho_v(1+i_v)s_v)^{\alpha_i}\right]^{1-\theta} - \left[\rho_d(1+i_d)\right]^{\alpha_i(1-\theta)} > \tau_i\left(F_i^v - F_i^d\right), \tag{10}$$

where $\tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}$. See Appendix B for the derivation of Equation (10).

From inspecting Equation (10) and assuming that the exchange rate s_v is log-normally distributed with mean μ and variance Σ, Appendix B proves the following result:

Proposition 1. Higher borrowing costs in vehicle currency, i_v, lowers the probability of invoicing in vehicle currency, ceteris paribus.

Invoicing imported inputs in vehicle currency v is optimal when its financing costs are sufficiently low to compensate for exchange rate uncertainty, after accounting for relative input costs and fixed costs of invoicing. Firms maintain a constant markup over marginal costs, so vehicle-currency invoicing implies that ex-post marginal costs depend on the realized exchange rate, potentially generating costly deviations from the optimal markup. Consequently, firms choose vehicle-currency invoicing only when its financing cost advantage outweighs the

extended to multi-market firms, capturing both within- and across-firm externalities.

risk of markup misalignment. While log-normality allows for closed-form solutions, the same intuition holds using a second-order approximation with a general distribution.

In addition, Appendix B proves that the trade-off between cheaper financing option and costly departure from the optimal markup depends on the exposure of the marginal cost to exchange rates fluctuations, α_i, capturing firm's operational hedging motive (Amiti et al., 2022):

Proposition 2. For sufficiently high exchange rate volatility, an increase in the cost of financing in vehicle currency, i_v, reduces the probability that a firm invoices in vehicle currency. This effect is stronger for firms with a higher share of imported inputs, α_i.

This proposition reflects the natural hedging behavior of firms in our setup. Since all revenues are assumed to be in domestic currency, firms optimally prefer to match the currency of their imported-input costs with that of their revenues. The incentive to do so is stronger the larger the share of imported inputs, α_i, because these firms are more exposed to fluctuations in vehicle-currency financing costs and deviations from the profit-maximizing price are more costly for them. Consequently, when i_v rises, the probability of invoicing in vehicle currency declines more sharply for firms with higher α_i.[12]

Lastly, Appendix B proves the following on the role of strategic complementarities:

Proposition 3. Stronger strategic complementarities – that is, lower fixed costs of invoicing – reduce the probability that a firm switches to domestic-currency invoicing when the cost of financing in vehicle currency, i_v, rises.

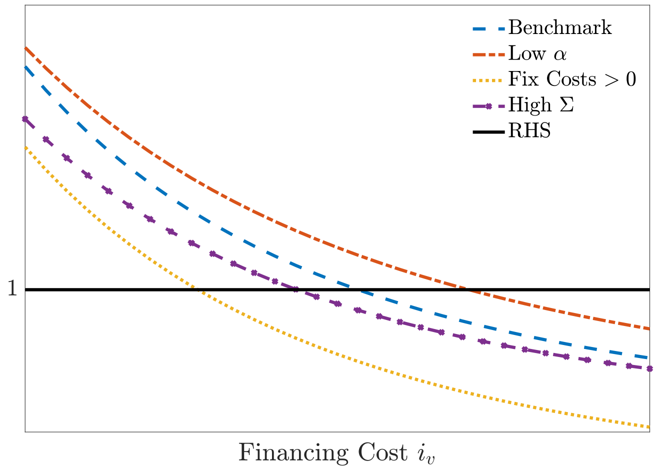
The term on the right-hand side of Equation (10) captures how fixed costs affect the invoicing choice.[13] Firms optimally invoice in vehicle currency only when the financing cost advantage is large enough to offset both the deviation from the optimal constant markup and the differential in fixed costs of invoicing. Thus, lower fixed costs of invoicing in vehicle currency, F_v, influenced by both the intensity of strategic complementarities in invoicing and by sunk costs, make vehicle currency borrowing more convenient even for higher levels of i_v.

Figure 3 graphically represents the optimal decision rule in Equation (10) and illustrates how it depends on the firm-specific share of intermediate inputs in production α_i, exchange rate volatility Σ, and the differential in currency-specific fixed costs $\left(F_i^v - F_i^d\right)$. Expected profits under domestic currency financing are normalized to one and independent of i_v (hori-

[12]Two distinct objects are at work. The first is the firm's financing need in vehicle currency, the working capital it must borrow and repay in that currency, which is what α_i measures. The second is the natural hedge against that need, the revenue the firm earns in the same currency. In the baseline model the second is zero by assumption, so α_i measures both the financing need and its unhedged part. Appendix C shows the opposite polar case: when revenues are denominated in vehicle currency, the same cutoff condition holds but the hedge lowers the effective threshold, so a larger shock is required to induce switching.

[13]In deriving the optimal invoicing choice, we assume that firms do not internalize the effect of their choice on the aggregate invoicing share and, consequently, on fixed costs.

Figure 3: Invoicing Choice - Comparative Statics



Note: Figure 3 plots the optimality condition in Equation (10), $\mathbb{E}\left[(\rho_v(1+i_v)s_v)^{\alpha_i}\right]^{1-\theta} - \tau_i\left(F_i^v - F_i^d\right) > [\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, for different levels of the vehicle currency interest rate i_v, assuming $s_v \sim$ Log-Normal(μ, Σ). The RHS, $[\rho_d(1+i_d)]^{\alpha_i(1-\theta)}$, is normalized to one. The blue line shows the benchmark calibration ($w = 0.2$, $\rho_v = \rho_d = 1$, $\theta = 5$, $\mu = 0$, $\Sigma = 1$, $\alpha_i = 0.8$, and $F_i^v - F_i^d = 0$). The other lines illustrate comparative statics relative to the benchmark: lower input share $\alpha_i = 0.7$ (red), higher fixed-cost differential $F_i^v - F_i^d = 1$ (yellow), and higher exchange rate variance $\Sigma = 1.05$ (purple).

zontal line). By contrast, expected profits when invoicing working capital in vehicle currency decline with i_v, since higher vehicle interest rates raise marginal costs (downward-sloping line). The intersection of the two curves determines the cut-off value of i_v above which it is no longer optimal to invoice in vehicle currency, that is, $\eta^\star = 1$.

Consistent with Proposition 2, a lower share of imported inputs, α_i, shifts the expected-profit curve for vehicle-currency invoicing to the right, increasing the likelihood that firms choose vehicle currency. In line with Proposition 3, a larger fixed-cost differential, $F_i^v - F_i^d$, shifts the curve leftward, making vehicle-currency invoicing less attractive. Finally, higher exchange rate volatility, Σ, raises the cost of deviating from the optimal markup, shifting the vehicle-currency expected-profit curve leftward and further reducing the probability that firms invoice in vehicle currency.

Lastly, to further analyze the mechanism, we can further assume that fixed costs are equal across currencies, $F_i^v = F_i^d$ for all i. Under this additional assumption, the cutoff condition can be rewritten as:

$$\eta_i^\star = 0 \qquad \text{iff} \qquad \underbrace{i_v - i_d + [\mu + 0.5\,\alpha_i \Sigma]}_{\text{UIP Deviation}} < \log\left(\frac{\rho_d}{\rho_v}\right), \tag{11}$$

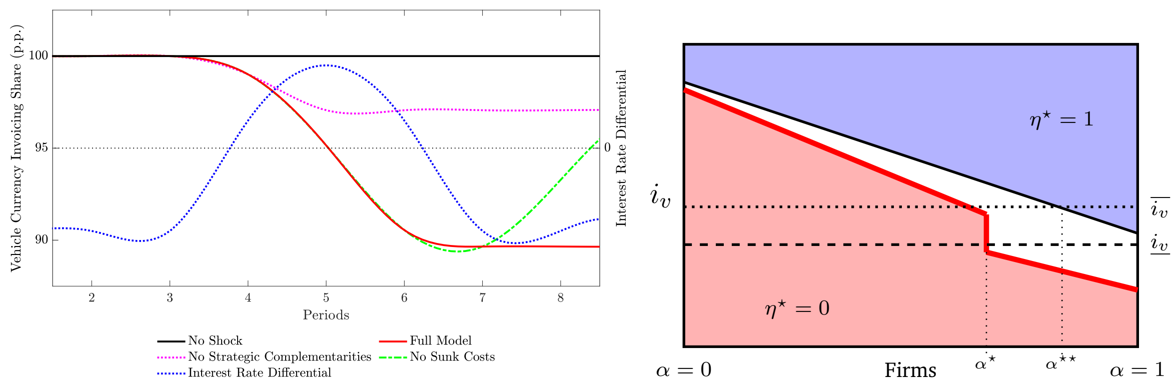
where μ and Σ are the mean and variance of $s_v \sim$ Log-Normal(μ, Σ). Equation (11) offers additional insights from the model: firm-specific UIP deviations shape importers' currency choices. The left hand side of Equation (11) shows that we can recast the key predictions of the models in terms of UIP fluctuations, drawing a direct link between UIP deviations and firm-level invoicing decisions and, in the aggregate, the dominance of a currency.

4.3 Temporary Shocks, Invoicing Dynamics and Hysteresis

Next, we qualitatively assess the effects of a temporary rise in the cost of finance in vehicle currency i_v – or, equivalently, an increase in UIP deviations – that permanently reverts to its original level after a few periods. We show that our theory can rationalize the main empirical findings of Section 3: the long lasting effects on the individual and aggregate import invoicing patterns. Moreover, firms that are larger and whose vehicle-currency financing needs are larger relative to their revenues are the first to switch to domestic currency invoicing. Figure 4 displays these dynamics.

The firm-level invoicing patterns presented in Section 3 support the assumption that all firms initially invoice in vehicle currency and have no prior experience invoicing in domestic currency. As a result, firms must pay a sunk cost to switch to domestic currency invoicing. Moreover, we assume that the share of imported goods used in production, α_i, is uniformly distributed: $\alpha_i \sim U[0, 1]$.[14]

Figure 4: Shock to Vehicle Currency Financing



Note: The right panel plots the optimal invoicing decision for each level of i_v (y-axis) conditional on the level of α (x-axis). The left panel plots the dynamics of the aggregate shares of US dollar invoicing. The red line represents the full model; the green one abstracts away from sunk costs; the purple one abstracts away from strategic complementarities. The blue line represents the dynamics of the relative costs of financing $i_v - i_d$. We use the following symbolic calibration: we normalize ρ_v and ρ_d to one; the elasticity of demand θ is set equal to 5; domestic wage is set to 0.2 (Saravia and Voigtländer, 2012); α_i are uniformly distributed; without loss of generality, we consider the following specification for the fixed costs of invoicing in US dollar: $F^v - F^d = k_0^v - f^d - \gamma^v \omega_{t-1}^v - \gamma_1^v(\omega_{t-2}^v - \omega_{t-1}^v)$, where γ^v is set equal to 0.7, $\gamma^d = 0$, $\gamma_1 = 2.5$, $k_0^v = 1$, and $f^d = \{k_1^d, k_0^d\} = \{0.2, 0\}$; we consider a 3% increase in $i_v - i_d$.

Short-run response The right panel of Figure 4 graphically represents each firm's invoicing choice. On the horizontal axis, firms are ranked in an ascending order based on their share of imported inputs, α_i. The vertical axis represents the cost of financing in vehicle currency, i_v. For any given level of i_v, there is a threshold $\tilde{\alpha}$ such that firms with $\alpha_i < \tilde{\alpha}$ invoice in domestic currency ($\eta^\star = 1$), while firms with $\alpha_i > \tilde{\alpha}$ invoice in vehicle currency ($\eta^\star = 0$). The downward-sloped line dividing the box in two areas – domestic (blue area) and vehicle (non-blue area) currency financing – represents the locus of $\tilde{\alpha}$ as a function of i_v.[15]

[14] The main insights are qualitatively unchanged regardless of the distributional assumption on α_i.

[15] For instance, in the special case where $s_v \sim$ Log-Normal(μ, Σ), the cutoff $\tilde{\alpha}$ is determined by Equation (11) once we solve explicitly for α_i.

Following the rise in the cost of financing in vehicle currency, larger firms and firms with larger unhedged vehicle-currency financing needs are the first to switch to domestic invoicing. The initial cost of financing in vehicle currency, $\underline{i_v}$, is low enough that it is optimal for all firms to invoice in vehicle currency. This is represented by $\underline{i_v}$ (dash horizontal line) lying below the initial locus of $\tilde{\alpha}$ (solid black line). When i_v increases from the initial level $\underline{i_v}$ to a higher level $\overline{i_v}$ – meaning foreign invoicing becomes less convenient – a mass of firms with $\alpha_i > \alpha^{\star\star}$ begin borrowing and invoicing in local currency. Notably, the first firms to switch are those with high α_i, which represents a sufficient statistics for both their unhedged exposure to vehicle currency and their market share in the imported inputs market.[16]

The presence of strategic complementarities creates a positive feedback in domestic currency invoicing, further expanding its use in imports. The left panel of Figure 4 shows that the aggregate share of imports invoiced in vehicle currency continues to decline even after the cost of financing in vehicle currency, i_v, starts to revert. The use of domestic currency by a group of firms after the shock reduces the fixed cost of invoicing in domestic currency for others, increasing the likelihood that more firms will adopt it. Firms with lower α_i value, relative to the initial switchers, begin to use domestic currency to invoice their imported inputs as domestic invoicing becomes more convenient for them.[17]

Hysteresis in invoicing The presence of sunk cost in invoicing implies that once a firm switches to domestic currency invoicing, the relative convenience of domestic versus vehicle currency invoicing is permanently altered. As a result, the sunk-cost model generates invoicing hysteresis: even if i_v permanently reverts to the original level $\underline{i_v}$, firms do not switch back to vehicle currency invoicing. Specifically, the difference between the fixed costs of invoicing in vehicle and domestic currency, $F^v - F^d$ in Equation (10), decreases, making domestic currency invoicing permanently more attractive than vehicle currency invoicing.

Graphically, this is represented by a downward shift in the threshold between domestic and vehicle currency invoicing for firms that begin invoicing in domestic currency (i.e. those with $\alpha_i > \alpha^\star$). The area where domestic invoicing is optimal ($\eta^\star = 1$) extends to include the area above the solid red line (i.e. both the white area and the original blue area). Thus, the new cut-off rule implies that for firms with $\alpha_i > \alpha^\star$, it remains optimal to invoice in domestic currency even when ϵ reverts to the original level, $\underline{i_v}$.[18]

At the aggregate level, this implies that the share of imports invoiced in vehicle currency

[16]The former follows immediately from the fact that firms are importers only, with revenues denominated in domestic currency. The latter follows from the fact that firms are assumed to have identical size. Thus, higher α_i implies more imported inputs and, thus, a larger market share in the input market.

[17]In the right panel of Figure 4, firms with $\alpha^\star < \alpha_i < \alpha^{\star\star}$ gradually switch to domestic currency invoicing as fixed costs decrease. At the aggregate level, the pink dashed line in the left panel shows that, when the i_v starts to revert, the share of vehicle currency invoicing stops decreasing in the absence of strategic complementarities.

[18]Notice that domestic currency invoicing becomes more appealing even for firms that have not yet paid the sunk cost, i.e. firms with $\alpha_i < \alpha^\star$. This is because the fixed cost of invoicing in domestic currency decreases also for them, due to strategic complementarities. Relative to the initial steady state, where all invoicing was in vehicle currency, the fixed cost of invoicing in vehicle currency is now higher since a subset of firms has permanently switched to domestic invoicing. The higher fixed cost of invoicing in vehicle currency F^v makes domestic invoicing more competitive for all firms.

does not return to one after i_v reverts to the original level, which is consistent with the observed data. The absence of any partial reversion in the aggregate share of imports invoiced in domestic currency suggests that the sunk cost associated with domestic currency invoicing is significant enough to make the switch permanent. The green dashed line in the left panel shows that, if the sunk cost were smaller, the aggregate share of invoicing could (partially) increase back, as it might become optimal for some firms to revert to vehicle currency invoicing. This would occur if the decrease in the fixed costs of domestic invoicing, F^d, is insufficient to make domestic currency invoicing optimal at the original level of i_v, $\underline{i_v}$. [19]

5 Model Validation

In this section we take the model to the data. First, we estimate a cross-sectional specification that tests the three propositions jointly: the probability of peso invoicing rises with the relative cost of dollar funding (Proposition 1), by more for firms whose dollar financing needs are less naturally hedged by dollar-invoiced export revenues (Proposition 2), and by less for firms whose competitors invoice in dollars (Proposition 3). We also perform event studies around the crisis and we recover the same two last margins in the time dimension. We next document the mechanism that makes the shift permanent, with three tests of the sunk cost of adopting a currency, and test the financing channel directly using firms' pre-crisis reliance on trade finance.

5.1 US Dollar-Chilean Peso UIP Dynamics

Following the theoretical framework, we construct realized dollar–peso UIP deviations as a time-varying proxy for the financing-cost differential between local and vehicle currencies, which we use to validate the model's predictions.[20] Specifically, we define the realized log UIP deviation between the Chilean peso and the US dollar as

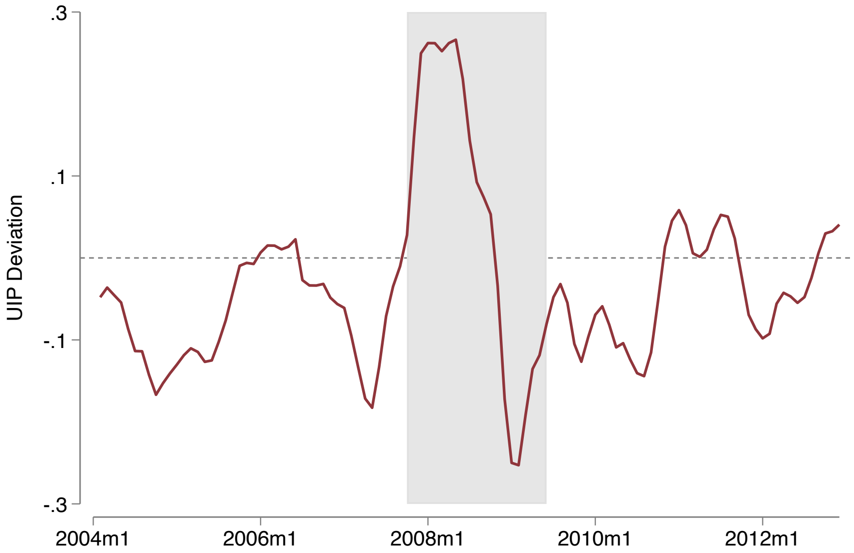
$$\text{UIP}_{t+1}^{\$,\text{CLP}} = \underbrace{i_t^{\$} - i_t^{\text{CLP}}}_{\Delta\text{Interest Rates}} + \underbrace{s_{t+1} - s_t}_{\Delta\text{Exchange Rate}} , \tag{12}$$

where $i_t^{\$} \approx \log(1 + i_t^{\$})$ and $i_t^{\text{CLP}} \approx \log(1 + i_t^{\text{CLP}})$ denote domestically-originated commercial lending rates on short-term (less than one year) loans denominated in US dollars and Chilean pesos, respectively. We use data on short-term bank lending rates in both currencies from

[19] Appendix C presents an exporter version of the model in which firms sell exclusively abroad and invoice revenues in US dollars. All three propositions carry over unchanged, since the currency-choice cutoff condition is identical to Equation (11). The key quantitative difference is that the natural hedge from dollar revenues is stronger, making vehicle-currency invoicing more attractive in the steady state. Thus, exporters require a larger UIP shock to switch permanently. Using the same calibration, we show that, in line with the empirical patterns, the same UIP shocks that moves invoicing patterns on the importer side does not have any long-lasting effect on the exporter side.

[20] In Figure 16 in Appendix A.6, we show that when the UIP deviation in Equation (12) is constructed using the one-year-ahead forecast exchange rate, the two measures display qualitatively similar dynamics. However, we used realized UIP in the empirical analysis because of data limitations.

Figure 5: Realized UIP Between the US Dollar and the Chilean Peso



Note: The Figure plots the monthly one-year realized UIP deviation between the Chilean peso and the US dollar, constructed using Equation (12). A positive value indicates that borrowing in Chilean pesos is relatively cheaper than borrowing in US dollars. Interest rates are measured using commercial lending rates on short-term (less than one year) loans denominated in Chilean pesos and US dollars offered by domestic banks. Data are monthly and smoothed using a two-month rolling window.

the Chilean Financial Authority responsible for supervising the banking system. These rates capture the interest cost faced by a domestic firm borrowing from domestic banks, either in Chilean pesos or in US dollars. The data cover the period around the GFC and are available until 2014. Because s_{t+1} is the realized exchange rate one year ahead, our UIP measure ends in 2013. Here, s_t denotes the log bilateral spot exchange rate (CLP per USD), and s_{t+1} is the realized exchange rate one year ahead. A negative UIP deviation ($\text{UIP}_{t+1} < 0$) indicates that borrowing in local currency is more costly than borrowing in US dollars, while a positive deviation implies the opposite.

During the GFC, borrowing in Chilean pesos became temporarily cheaper than borrowing in US dollars, reversing the pattern observed in normal times. Figure 5 shows that outside the peak of the crisis, realized UIP deviations are negative, fluctuating around -10 percentage points, indicating that borrowing in local currency is typically more expensive than borrowing in US dollars.[21] In contrast, during the GFC the average realized UIP deviation increased sharply, turned positive, and peaked at nearly 30 percentage points, implying a sudden rise in the cost of dollar-denominated borrowing. This pattern is consistent with a shortage of dollar funding at the height of the financial crisis (Ahn et al., 2011; Amiti and Weinstein, 2011).[22] By 2009, the relative cost of financing had reverted to its negative, pre-crisis level.[23] The transitory nature of the shock to the relative cost of financing in local and vehicle currencies

[21]The magnitude is consistent with existing evidence in the literature; see Kalemli-Ozcan and Varela (2021).

[22]Figure 17 in Appendix A plots the share of domestic bank credit to firms denominated in Chilean pesos, at monthly frequency. The share rises by roughly two percentage points between 2007 and 2009, remains elevated through 2010, and returns to its pre-crisis level from 2011.

[23]These dynamics are in line with previous evidence on currency-specific borrowing costs in Chile; see Vial et al. (2020) and Betancour et al. (2006).

is consistent with the theoretical mechanism required to generate a persistent decline in the share of vehicle-currency invoicing. Importantly, the timing and magnitude of these movements do not coincide with changes in banking or capital-market regulation. Major reforms to liberalize the financial sector, enhance supervision, and deepen international integration, such as improvements in access to foreign-exchange hedging instruments, were implemented only after the crisis period (IMF, 2010). Similarly, Chile adopted key prudential regulations later than other advanced and OECD economies.[24]

5.2 Cross-Sectional Evidence

Using cross-sectional variation, we test the key predictions of the theoretical framework. We start by estimating the following specification:

$$\begin{aligned} Peso_{fpot} = \beta_0 + \beta_1 \text{UIP}_t^{\$,\text{CLP}} + \beta_2 \text{NH}_{ft}^{\$} + \beta_3 \text{SC}_{(-f)ost}^{\$} + \\ + \beta_4 \text{NH}_{ft}^{\$} \times \text{UIP}_t^{\$,\text{CLP}} + \beta_5 \text{SC}_{(-f)ost}^{\$} \times \text{UIP}_t^{\$,\text{CLP}} + \\ + FE_{fpo} + \nu_{fpot}, \end{aligned} \tag{13}$$

where the dependent variable is a dummy variable that takes value equal one when the transaction at the firm-origin-product(HS8)-time ($fopt$) level is invoiced in Chilean peso and zero if in US dollar. $\text{UIP}_t^{\$,\text{CLP}}$ is the constructed following Equation (12) and represented in Figure 5, proxying the relative cost of financing in local and vehicle currency. $\text{NH}_{ft}^{\$}$ and its interaction with UIP deviations capture the key prediction of Proposition 2. The prediction relates two distinct objects. The first is the firm's financing need in US dollars (α_i). The second is the natural hedge against that need: the revenue the firm earns in dollars which offsets it. What governs the invoicing decision is the difference between the two. We measure the hedge directly, as the share of firm f's exports invoiced in US dollars at time t, $\text{NH}_{ft}^{\$}$, and use it as an inverse proxy for the unhedged need. An importer that already earns a large share of its revenue in dollars is naturally hedged and has less to gain from moving its import invoicing out of the dollar when dollar funding becomes expensive, so Proposition 2 predicts a negative coefficient on $\text{NH}_{ft}^{\$} \times \text{UIP}_t^{\$,\text{CLP}}$.[25] Lastly, the term $\text{SC}_{(-f)ost}^{\$}$ captures strategic complementarities in invoicing. We measure strategic complementarities in invoicing as in Crowley et al. (2020):

$$\text{SC}_{(-f)ost}^{\$} = \frac{\sum_{k \neq f} \text{ToT Imp}_{kost}^{\$}}{\sum_c \sum_{k \neq f} \text{ToT Imp}_{kost}^{c}} \tag{14}$$

[24]See Madeira and Olivares (2021) and https://www.sipa.columbia.edu/sites/default/files/2023-02/Macroprudential-Policies-A-View-From-Chile.pdf. Basel III was implemented in 2019; see https://www.cmfchile.cl/portal/principal/613/w3-article-50324.html.

[25]α_i has a direct counterpart in the data, the firm's dollar import share, but it cannot enter Equation (13): the dependent variable is an indicator for the same transaction being invoiced in pesos, and the dollar import share is a value-weighted aggregate of that indicator over the same firm and period, so a firm that switches a transaction to the peso mechanically lowers its own regressor. The import-side measure is used only in the event study of Equation (17), where it is fixed before the crisis and hence predetermined.

where $(-f)$ is the set of competitors of firm f in sector s (HS4) who import from origin o at time t. Equation (14) measures the share of total imports of competitors at the industry (HS4) and origin level invoiced in US dollar. Clearly, the invoicing choice of firm f is endogenous to the choice of its competitors $(-f)$. Following previous work (Amiti et al., 2022; Crowley et al., 2020), we construct the following instrumental variables to correct for the existence of possible endogeneity:

$$\widehat{\text{SC}}^{\$}_{(-f)ost} = \sum_{k \neq f} \frac{S_{kost}}{1 - S_{fost}} \times \frac{\text{ToT Exp}^{\$}_{kt}}{\sum_c \text{ToT Exp}^{c}_{kt}}, \tag{15}$$

where $S_{kost} = \frac{\text{ToT Imp}_{kost}}{\sum_f \text{ToT Imp}_{fost}}$ is therefore the import market share of firm k from origin o, sector s (HS4), and time t level among all Chilean importers, and the last ratio is the share of firm k's total exports that are invoiced in US dollars. Following Crowley et al. (2020), we complement this instrument with a second leave-out shifter built on the same import-share weights, replacing competitors' US dollar export share with their log total export value:

$$\widehat{\text{TOTEXP}}_{(-f)ost} = \sum_{k \neq f} \frac{S_{kost}}{1 - S_{fost}} \times \log\left(\sum_c \text{ToT Exp}^{c}_{kt}\right). \tag{16}$$

The first shifter moves competitors' currency mix, the second their scale. We instrument $\text{SC}^{\$}_{(-f)ost}$ and its interaction with UIP deviations using both shifters and their interactions with $\text{UIP}^{\$,\text{CLP}}_t$.

The specification in Equation (13) allows us to empirically test the key proposition of the model by exploiting cross-sectional variation. Proposition 1 predicts that an increase in borrowing costs denominated in vehicle currency raises the probability that imports are invoiced in local currency, implying a positive coefficient β_1. Proposition 2 predicts a negative coefficient for β_4. When dollar borrowing becomes relatively more expensive, the firms with the strongest incentive to move their import invoicing to the peso are those whose dollar financing needs are least offset by dollar revenues. A higher share of exports invoiced in US dollars means a larger natural hedge and hence a smaller unhedged need, so it is associated with a weaker response to the same movement in relative funding costs, and β_4 is expected to be negative. Proposition 3 further predicts that the effect of changes in vehicle-currency borrowing costs is weaker in the presence of stronger strategic complementarities in invoicing. Accordingly, β_5 is expected to be negative.

Table 2 reports the estimates from Equation (13) and shows that the data support the key predictions of the model. Consistent with Proposition 1, Columns (1) - (3) show that the coefficient on the UIP deviations, β_1, is positive and significant. A one-percentage point increase in the cost of financing in US dollar, proxied by the UIP deviation, raises the probability of invoicing in Chilean peso by approximately 2.7 basis points in Columns (1) and (2). In Columns (3) and (4), which include the interaction terms, the marginal effect evaluated at the sample means of $\text{SC}^{\$}_{(-f)ost}$ and $\text{NH}^{\$}_{ft}$ is 2.5 basis points. This result aligns with the

Table 2: Model Validation: Cross-Sectional Analysis

	OLS			IV
	(1)	(2)	(3)	(4)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
$UIP_t^{\$,PESO}$	0.027***	0.026***	0.153***	0.188***
	(0.001)	(0.001)	(0.013)	(0.025)
$SC_{kt}^{\$}$		-0.005***	-0.006***	-0.023**
		(0.001)	(0.001)	(0.009)
$NH_{ft}^{\$}$		-0.012***	-0.013***	-0.012***
		(0.001)	(0.001)	(0.001)
$SC_{kt}^{\$} \times UIP_t^{\$,PESO}$			-0.049***	-0.093***
			(0.006)	(0.029)
$NH_{ft}^{\$} \times UIP_t^{\$,PESO}$			-0.087***	-0.084***
			(0.011)	(0.012)
Firm-Origin-HS8	Yes	Yes	Yes	Yes
Weak IV F-Stat				312.38
Hansen J-stat				2.177
Hansen p-value				0.337
Observations	1075005	1054967	1054967	1054967

Note: The Table reports the coefficients from specification in Equation (13). The dependent variable is a dummy variable equal to 1 when the $fopt$ transaction is invoiced in Chilean peso and 0 in US dollar. Column (3) estimates Equation (13) by OLS. Column (4) instruments $SC_{kt}^{\$}$ with two leave-one-out, import-share-weighted averages of competitors' characteristics in the same sector(HS4)-origin-year market: competitors' US dollar export share and competitors' log total export value. The interaction $SC_{kt}^{\$} \times UIP_t^{\$,PESO}$ is instrumented by the corresponding interactions of these two averages with $UIP_t^{\$,PESO}$. All Columns except Column (1) control for importer size (firm f's share of aggregate imports). Standard errors are clustered at the firm-product-origin level. We use data from Garcia-Marin et al. (2019) for 2004–2013, the period for which our UIP measure is available. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

model's prediction that an increase in the expected cost of borrowing in US dollar relative to the Chilean peso strengthens firms' incentives to invoice in domestic currency. Given the roughly 40-percentage-point increase in our UIP measure during the GFC (Figure 5), this implies that the probability of invoicing in local currency increased by approximately one percentage point, accounting for roughly half of the aggregate dynamics shown in Figure 2. Appendix A.7 shows that a similar UIP coefficient is obtained when the sample is restricted to 2004–2007, before the GFC, confirming that the UIP-invoicing relationship is a stable structural feature of the data rather than an artifact of the crisis episode.

The main specification in Column (4) provides evidence in line with Propositions 2 and 3. The interaction terms between the borrowing-cost differential and the natural hedge ($\mathrm{NH}_{ft}^{\$}$) and strategic complementarities in invoicing ($\mathrm{SC}_{(-f)ost}^{\$}$) are both negative and precisely estimated. These results suggest that, when US dollar borrowing costs rise, firms are less likely to switch to local-currency invoicing if their competitors invoice a large share of imports in dollars or if they themselves invoice a large share of exports in dollars. In line with the theoretical framework, strategic complementarities in vehicle-currency invoicing raise the effective cost of moving away from it, dampening the incentive to adopt the peso. Similarly, a firm that already earns a large share of its revenue in dollars so its unhedged dollar financing need is smaller and it has less reason to re-denominate its imports when dollar funding becomes expensive.

Finally, the same cross-sectional determinants govern not only the level of peso invoicing but the switching margin itself. Table 16 in Appendix A.9 re-estimates Equation (13) replacing the invoicing indicator with the probability of switching from US dollar to peso invoicing. An increase in the relative cost of dollar funding raises the probability of switching, and the interactions with natural hedging and strategic complementarities retain the negative signs of Table 2. The same determinants also predict the order in which firms adopt. Figure 20 in Appendix A.10 plots the cumulative share of firms that have adopted peso invoicing by each year between 2008 and 2012, by pre-crisis exposure group: adoption is monotone in the strategic complementarity of the firm's markets, with firms in the least dollar-dominated markets adopting first and firms in the most dollarized markets adopting last, in line with the ranking of Proposition 3, while fully hedged exporters adopt at roughly half the rate of partially hedged exporters throughout, consistent with Proposition 2.

5.3 Event Study Design

We provide additional econometric evidence on the heterogeneous effects due to the dollar financing need and strategic complementarities using event study specifications. We estimate the following specification:

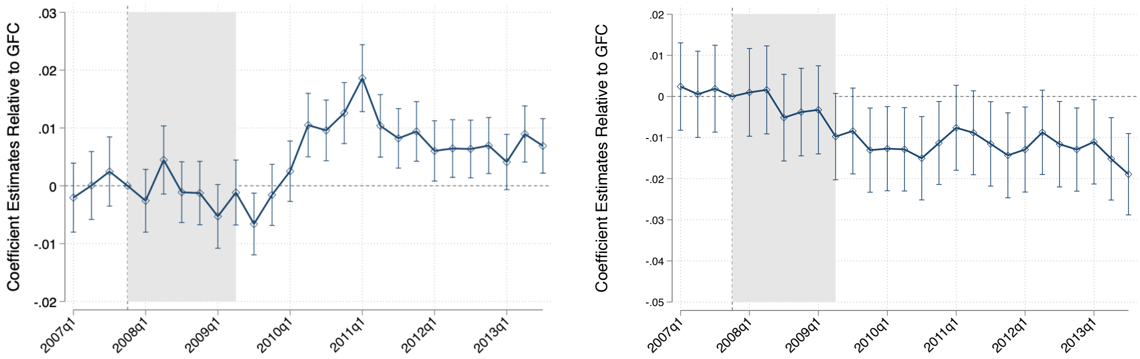
$$Peso_{fopt} = \sum_{s} \beta_s \left(X_f \times \mathbf{1}[t = s]\right) + FE_{fpo} + FE_t + \epsilon_{fpot}, \tag{17}$$

where the dependent variable is still a dummy variable that takes value equal one when the transaction at the firm-origin-product(HS8)-time level is invoiced in Chilean peso and zero if in US dollar; $\mathbf{1}[t = s]$ is an indicator equal to 1 in quarter s and 0 otherwise. The sample runs from 2007Q1 to 2017Q4 and all coefficients are reported relative to 2007Q4, the quarter in which the crisis begins.

The term X_f is the firm-level exposure, fixed before the crisis. When assessing the role of the dollar financing need (Proposition 2), X_f is the share of firm f's imports denominated in US dollar, averaged over the quarters in which the firm is active between 2007Q1 and 2007Q3.[26] When assessing the role of strategic complementarities (Proposition 3), X_f is the leave-one-out share of competitors' imports invoiced in US dollar in the same origin-sector(HS4) market, averaged over the same window as in Equation (14). Because both measures are constructed from invoicing decisions taken before the shock, they are not affected by firms' subsequent invoicing choices, which avoids the reverse-causality concerns that may arise in the cross-sectional regressions.

[26] Note that the event study measures the dollar financing need directly, on the import side, whereas the cross-sectional specification in Equation (13) measures the natural hedge against that need on the export side, $\mathrm{NH}^{\$}_{ft}$, and therefore enters as an inverse proxy. The two designs consequently predict coefficients of opposite sign for the same mechanism.

Figure 6: Event Study Design



Note: The Figure plots the coefficients β_s from the event study specification in Equation (17). The left panel measures the firm's dollar financing need (Proposition 2), proxied by the share of imports denominated in US dollar over 2007Q1–2007Q3. The right panel uses the leave-one-out strategic complementarity measure of Equation (14), constructed over the same window (Proposition 3). Both panels absorb firm-product-origin and time fixed effects. The coefficient in 2007Q4 is normalized to zero and marked by the vertical dashed line; the shaded area covers 2007Q4–2009Q2. Data are from the Chilean Customs Agency; the specification is estimated through 2017Q4 and coefficients are displayed through 2013Q4. We report 95 percent confidence intervals. Standard errors are clustered at the firm-product-origin level.

Identification comes from interacting time indicators with firms' pre-crisis exposure, while saturating the specification with firm-product-origin and time fixed effects. Accordingly, β_s measures the differential change at quarter s in the probability of invoicing in Chilean peso between a firm with full pre-crisis exposure ($X_f = 1$) and an otherwise identical firm with no exposure ($X_f = 0$), relative to 2007Q4.

The theory predicts that, following an increase in the cost of borrowing in US dollars, firms with a larger pre-GFC dollar financing need (larger share of imports denominated in US dollars) are more likely to switch to domestic currency invoicing. Consistent with Proposition 2, this implies positive values of β_s. In contrast, firms operating in environments characterized by stronger strategic complementarities in US dollar invoicing face higher effective switching costs and are therefore less likely to move toward peso invoicing when dollar financing costs rise. In line with Proposition 3, this mechanism translates into negative values of β_s.

The left panel of Figure 6 shows the financing need margin. Over the three quarters preceding the crisis the coefficients are small and statistically indistinguishable from zero, and they remain so throughout the crisis window itself. The positive differential, as predicted by Proposition 2, emerges in the course of 2010, rises quickly to a peak of roughly 1.9 percentage points in 2011Q1, and remains flat thereafter. The magnitude at the peak is large relative to the unconditional probability of peso invoicing in this panel, which is below one percent. The right panel reports the same design using the strategic complementarity index. The coefficients are negative, as Proposition 3 predicts, from 2010 onward and reach roughly -2 percentage points by 2013. The path, however, declines steadily over the whole sample rather than breaking at the crisis, so we read this panel as documenting that markets dominated by dollar-invoicing competitors adopt the peso more slowly, rather than as identifying a response timed to the GFC.

In Appendix A.6, we show that the qualitative results hold under alternative fixed-effect structures. Figure 18 replaces the time fixed effects on the financing need margin with origin-time and with origin-product-time effects, and Figure 19 replaces them on the strategic complementarity margin with sector-time and with firm-time effects.

5.4 Sunk Costs in Invoicing

We provide evidence, consistent with the international trade literature, that supports the presence of sunk costs in invoicing decisions, which is the key mechanism explaining the hysteresis in currency choice in our framework. The key implication of sunk costs is that the decision to invoice in a specific currency is more likely if the same currency has been previously used. Therefore, conditional on other determinants of invoicing, the past invoicing share should predict current invoicing choices. This idea translates into the following econometric specification:

$$Peso_{fopt} = \beta y^{Peso}_{f,t-1} + Controls_{fopt} + FE_{fop} + FE_t + \nu_{fopt}, \tag{18}$$

where $Peso_{fopt}$ is a dummy variable equal to one when the transaction at the firm-origin-product-time level is invoiced in Chilean peso and zero if in US dollar, $y^{Peso}_{f,t-1}$ is the lagged share of imports invoiced in Chilean peso at the firm level, and β is the coefficient of interest.

Including the lagged endogenous variable raises potential endogeneity concerns, as serially correlated unobserved factors may induce persistence in invoicing choices and upwardly bias the coefficient β. Following the literature, we include a comprehensive set of fixed effects to mitigate these concerns (Bernard and Jensen, 2004; Timoshenko, 2015; Das et al., 2007). Moreover, we introduce additional controls affecting invoicing choices, such as natural hedging and strategic complementarities, defined as in Equation (13).

Table 3 shows that a higher lagged share of imports invoiced in local currency increases the probability of invoicing in Chilean pesos, consistent with the presence of sunk costs. The OLS specification in Column (1) indicates that a one-percentage-point increase in the lagged share of imports invoiced in local currency raises the probability of local-currency invoicing by 77 percentage points. The benchmark specification in Column (4), which includes fixed effects and additional controls, shows that a one-percentage-point increase in the lagged share increases the probability of local-currency invoicing by approximately 40 basis points. The smaller magnitude in the benchmark specification is consistent with an upward bias in the OLS estimates due to omitted fixed effects and controls.[27]

In the model, Equation (8) assumes that the firm-specific component of the fixed cost of invoicing falls permanently upon first use of a currency. We provide evidence consistent with

[27] A competing explanation is mechanical persistence: the realized UIP deviation is itself serially correlated, so under this alternative the lagged peso share coefficient should fall toward zero once UIP normalizes. Appendix A.8 distinguishes the sunk-cost interpretation from this serial-correlation alternative by interacting the lagged peso share with a post-2010 indicator, by which point the UIP deviation had returned to its pre-crisis level. The interaction is economically negligible and not significant at conventional levels: persistence is essentially unchanged after the UIP incentive disappears. Mechanical serial correlation would instead imply a substantial fall.

Table 3: Sunk Cost Mechanism Validation

	OLS			IV
	(1)	(2)	(3)	(4)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
Lagged CLP Share$_{f,t-1}$	0.7705***	0.4019***	0.3992***	0.4003***
	(0.0048)	(0.0065)	(0.0065)	(0.0065)
Controls	No	No	Yes	Yes
Firm-Origin-HS8 & Year	No	Yes	Yes	Yes
Observations	4142454	2923670	2883077	2882403

Note: The Table reports coefficients from specification in Equation (18). The dependent variable equals 1 when the $fopt$ transaction is invoiced in Chilean peso and 0 in US dollars. "Lagged CLP Share" refers to the lagged share of imports invoiced in local currency at the firm level. Controls include: i) a natural hedging measure defined as the share of exports invoiced in US dollars; ii) a strategic complementarities index defined as the average share of imports invoiced in US dollars among all competitors in a given sector(HS4)-origin pair as in Equation (14). IV columns refer to the case where the strategic complementarity index is instrumented with the import-weighted average US dollar exposure of all competitors in the sector(HS4)-origin pair as in Equation (15) and with the import-share weighted averages of competitors' log total export value as in Equation (**??**); iii) a dummy equal to 1 if the importer also exports and 0 otherwise. The sample covers 2003–2015 from Garcia-Marin et al. (2019). Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

this mechanism with the following specification:

$$Peso_{fopt} = \beta_1 E^{Peso}_{f,t-1} + \beta_2\, E^{Peso}_{f,t-1} \times \text{UIP}^{\$,\text{CLP}}_t + \beta_3\, \text{UIP}^{\$,\text{CLP}}_t + \\ + \delta\, y^{Peso}_{f,t-1} + Controls_{fopt} + FE_{fop} + FE_t + \nu_{fopt}, \quad (19)$$

where $Peso_{fopt}$ is a dummy variable equal to one when the transaction at the firm-origin-product-time level is invoiced in Chilean peso and zero if in US dollar, and $E^{Peso}_{f,t-1}$ is an indicator equal to one if firm f invoiced at least one import transaction in Chilean peso in any year prior to t. $\text{UIP}^{\$,\text{CLP}}_t$ is the realized UIP deviation defined in Equation (12), $y^{Peso}_{f,t-1}$ is the lagged share of imports invoiced in Chilean peso at the firm level as in Equation (18), and controls include the natural hedging and strategic complementarity measures defined above. The model predicts $\beta_1 > 0$: ever-adopters face the lower recurring cost, raising the probability of peso invoicing conditional on all other determinants. It also predicts $\beta_2 > 0$: a lower fixed-cost differential shifts the cutoff in Equation (10), placing past adopters closer to the switching margin and making their invoicing choice more responsive to the relative cost of dollar funding.

Table 4 shows that the data strongly support both predictions. Column (1) indicates that prior adoption raises the probability of invoicing in Chilean peso by 4.4 percentage points, an effect more than twenty times the 0.2 percent baseline probability among firms that have never used the peso. Column (2) shows that the estimate is unaffected by the natural hedging and strategic complementarity controls. Column (3) indicates that the effect does not reflect mechanical persistence in invoicing intensity: conditional on the lagged peso share of Equation (18), having ever adopted still raises the probability of peso invoicing by 2 percentage points, consistent with a discrete and permanent reduction in the cost of using

Table 4: Sunk Cost of Currency Adoption: Ever-Adopters vs. Never-Adopters

	(1)	(2)	(3)	(4)	(5)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
Ever CLP$_{f,t-1}$	0.0439***	0.0435***	0.0197***	0.0392***	0.0374***
	(0.0009)	(0.0009)	(0.0006)	(0.0010)	(0.0010)
Ever CLP$_{f,t-1}$ $\times$ $UIP_t^{\$,PESO}$				0.0245***	0.0232***
				(0.0052)	(0.0052)
$UIP_t^{\$,PESO}$				0.0045***	
				(0.0004)	
Lagged CLP Share$_{f,t-1}$			0.3798***		
			(0.0065)		
Firm-Origin-HS8	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	No	Yes
Controls	No	Yes	Yes	Yes	Yes
Observations	3108641	3066155	2883077	2451523	2451523

Note: The Table reports coefficients from Equation (19). The dependent variable equals 1 when the $fopt$ transaction is invoiced in Chilean peso and 0 in US dollar. "Ever CLP$_{f,t-1}$" is an indicator equal to one if firm f invoiced at least one import transaction in Chilean peso in any year strictly before t. "Lagged CLP Share$_{f,t-1}$" refers to the lagged share of imports invoiced in local currency at the firm level. $UIP_t^{\$,PESO}$ is the realized UIP deviation between the US dollar and the Chilean peso defined in Equation (12). Controls include: i) a natural hedging measure defined as the share of exports invoiced in US dollars together with an indicator for exporting firms; ii) a strategic complementarities index defined as the average share of imports invoiced in US dollars among all competitors in a given sector(HS4)-origin pair as in Equation (14) iii) a dummy equal to 1 if the importer also exports and 0 otherwise. The sample covers 2004–2015 from Garcia-Marin et al. (2019); columns (4) and (5) cover 2004–2013, the period for which UIP deviations are available. Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

the currency rather than with serially correlated invoicing behavior. Columns (4) and (5) show that the interaction between the ever-adopter indicator and the realized UIP deviation is positive and statistically significant: as predicted, past adopters respond more strongly to the same movement in relative funding costs. Given the roughly 40-percentage-point swing in our UIP measure during the GFC, the estimate implies that the peso-invoicing response of experienced firms exceeded that of inexperienced firms by about 1 percentage point.

The model interprets the sunk cost κ_0 as a firm $\times$ currency specific set-up cost. In other words, the sunk cost is assumed to be paid at the level of the firm, not of the individual trade relationship. Thus, experience acquired in one market could lower the cost of adopting the same currency in the firm's other markets (Crowley et al., 2020). We test this prediction on the panel of firm-origin-product combinations that have not yet adopted Chilean peso invoicing, restricting the sample to observations that had never invoiced imports in Chilean pesos up to $t1$ and excluding all years after the first adoption, using the following specification:

$$Adopt_{fopt} = \gamma\, E_{f,t-1}^{Peso,-op} + Controls_{ft} + FE_{fop} + FE_{ost} + \varepsilon_{fopt}, \tag{20}$$

where $Adopt_{fopt}$ is a dummy variable equal to one when a fop combination invoices imports in Chilean peso for the first time in year t, and $E_{f,t-1}^{Peso,-op}$ is an indicator equal to one if firm f invoiced at least one import transaction in Chilean peso before t in markets (origin $\times$ product)

Table 5: Within-Firm Spillover of Currency Adoption Across Markets

	(1)	(2)	(3)
	$Adopt_{fopt}$	$Adopt_{fopt}$	$Adopt_{fopt}$
Ever CLP other Markets$_{f,t-1}$	0.0280***	0.0320***	
	(0.0003)	(0.0006)	
Ever CLP other Origins$_{f,t-1}$			0.0327***
			(0.0006)
Firm-Origin-HS8	No	Yes	Yes
Origin-HS4-Year	Yes	Yes	Yes
Controls	No	Yes	Yes
Observations	5784468	3861411	3861411

Note: The Table reports coefficients from Equation (20). The sample consists of the panel of firm-product-origin triplets that have not yet adopted Chilean peso invoicing, that is, triplets that have never invoiced imports in Chilean pesos up to $t-1$. Observations after a triplet's first adoption are excluded. The dependent variable equals 1 when the triplet invoices imports in Chilean peso for the first time in year t and 0 otherwise. "Ever CLP other markets$_{f,t-1}$" is an indicator equal to one if firm f invoiced at least one import transaction in Chilean peso in any year strictly before t; within the sample of firm-product-origin triplets that have not yet adopted Chilean peso invoicing, such experience necessarily originates from the firm's other product-origin markets. "Ever CLP other origins$_{f,t-1}$" restricts experience to that acquired in markets located in a different origin country. Invoicing histories are observed from 2003, when currency information becomes available, so both experience measures are left-censored at 2003. Controls in Columns (2) and (3) include a natural hedging measure defined as the share of exports invoiced in US dollars together with an indicator for exporting firms. All columns include sector(HS4)-origin-year fixed effects, which absorb the strategic complementarities index of Equation (14) together with any other market-level determinant of invoicing, including the aggregate dynamics of UIP deviations; columns (2) and (3) additionally include firm-product-origin fixed effects. The sample covers 2004–2015 from Garcia-Marin et al. (2019). Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

other than op. The FE_{ost}, defined at the origin-sector(HS4)-year level, absorbs the strategic complementarity index of Equation (14) together with every other market-level determinant of invoicing, including the aggregate dynamics of UIP deviations. Thus, identification relies on comparing firms operating in the same market at the same time, differing only in the invoicing experience accumulated elsewhere. A positive γ indicates that adopting a new invoicing currency is the acquisition of a firm-level capability, the defining feature of the sunk cost κ_0, rather than a market-specific relationship, which would imply $\gamma = 0$.

Table 5 shows that spillovers are strong. Experience acquired in the firm's other markets raises the annual adoption probability by 2.8 to 3.2 percentage points, roughly a tenfold increase relative to the 0.3 percent baseline annual adoption rate among firm-origin-product combinations that have not yet adopted Chilean peso invoicing, and Column (2) shows that the estimate survives comparing the same firm-origin-product over time. In Column (3), experience is restricted to markets which are defined as origin countries other than o, and delivers an effect of the same size (3.3 percentage points). Together with Table 4, this evidence is consistent with the existence of a fixed cost mechanism that operates at the firm $\times$ currency level.

5.5 Trade Finance and the Switch to Peso

The theoretical framework assumes that all imports are financed through loans denominated in either the domestic or a vehicle currency. In practice, however, international trade is also commonly financed through trade credit, whereby the exporter allows the importer to defer payment (Benguria et al., 2026; Hardy et al., 2022). This differs from payment arrangements in which the importer bears the financing cost directly. Under a letter of credit, for example, the importer's bank pays the exporter against the shipping documents in the currency specified in the credit and extends the corresponding financing to the importer. Under cash in advance, by contrast, the importer pays before shipment using its own funds or borrowed resources. Both arrangements are forms of trade finance.

A rise in the relative cost of dollar funding should therefore induce importers that rely more heavily on trade finance to shift toward peso invoicing, as their financing needs are more binding. We test this prediction by examining whether invoicing behavior around the GFC differs systematically with firms' reliance on trade finance. Importantly, because payment terms and invoicing currency are jointly determined, we measure firms' reliance on trade finance prior to the crisis and then study subsequent changes in invoicing currency using the following empirical specification:

$$y_{fopt} \;=\; \beta\, X^{0407} + \gamma \left(X^{0407} \times \text{UIP}_t^{\$,\text{CLP}} \right) + FE_{ost} + \varepsilon_{fopt}, \tag{21}$$

where y_{fopt} is either $Switch_{fopt}$, equal to one when a firm-origin-product combination invoiced exclusively in US dollars at $t-1$ and invoices in Chilean peso at t, estimated on the combinations at risk of switching, or $Adopt_{fopt}$, equal to one in the year of first peso use, on the panel of combinations that have not yet adopted. The exposure X^{0407} is the share of 2004–2007 import value paid through trade finance (cash in advance, CIA, plus letters of credit, LC), measured at the combination (fop) or firm (f) level; trade credit (supplier credit) is instead the omitted category. The origin-sector(HS4)-year fixed effects absorb every market-level determinant of switching, including the level of UIP deviations, so identification relies exclusively on cross-sectional differences in pre-crisis financing dependence within the same market and year.

In line with our theoretical framework, Table 6 shows that importers more dependent on trade finance were substantially more likely to move to the peso when dollar funding became relatively expensive. The effect is present at the combination level (Columns 1–2), unaffected by the natural hedging and strategic complementarity controls (Column 2), and largest when dependence is measured at the firm level (Column 3), consistent with financing being a firm-wide margin and with the sunk cost operating at the firm-currency level, although less precisely estimated. Column 4 shows a consistent pattern using the lagged financing mix. Columns 5 and 7 split trade finance into its components: the response is strongest for relationships financed through letters of credit, the instrument in which dollar bank credit is bundled most tightly with dollar invoicing (the hypothesis $\gamma_{LC} = \gamma_{CIA}$ is rejected with $p < 0.01$ in

Table 6: Financing Dependence and the Switch to Peso Invoicing

	$Switch_{fopt}$					$Adopt_{fopt}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
TF^{0407}_{fop}	-0.0018***	-0.0016***				-0.0017***	
	(0.0002)	(0.0002)				(0.0001)	
$TF^{0407}_{fop} \times UIP^{\$,PESO}_t$	0.0127***	0.0130***				0.0239***	
	(0.0021)	(0.0021)				(0.0020)	
TF^{0407}_{f}			-0.0026**				
			(0.0010)				
$TF^{0407}_{f} \times UIP^{\$,PESO}_t$			0.0194*				
			(0.0109)				
$TF_{fop,t-1}$				-0.0019***			
				(0.0001)			
$TF_{fop,t-1} \times UIP^{\$,PESO}_t$				0.0054***			
				(0.0016)			
CIA^{0407}_{fop}					-0.0022***		-0.0019***
					(0.0002)		(0.0001)
LC^{0407}_{fop}					-0.0006*		-0.0010***
					(0.0003)		(0.0003)
$CIA^{0407}_{fop} \times UIP^{\$,PESO}_t$					0.0094***		0.0218***
					(0.0020)		(0.0020)
$LC^{0407}_{fop} \times UIP^{\$,PESO}_t$					0.0239***		0.0317***
					(0.0040)		(0.0035)
Origin-Sector-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	596383	595246	844722	952017	596383	799536	799536

Note: The Table reports coefficients from Equation (21). In Columns (1)–(5) the dependent variable equals 1 when a firm-product-origin combination invoiced imports exclusively in US dollars at $t-1$ and invoices in Chilean peso at t; the sample is the set of combinations at risk of switching, with mean 0.0024. In Columns (6)–(7) the dependent variable equals 1 in the year of the combination's first Chilean peso use, on the panel of combinations that have not yet adopted, excluding observations after first adoption, with mean 0.0033. Payment forms are classified as trade finance (cash in advance, CIA; letters of credit, LC) and trade credit (supplier credit, the omitted category); mixed and no-payment transactions are excluded. Exposures are shares of 2004–2007 import value, fixed before the crisis, at the firm-product-origin (fop) or firm (f) level; Column (4) uses the lagged share. Controls in Column (2): the natural hedging measure of Equation (13), an exporter indicator, and the strategic complementarity index of Equation (14). All columns include origin-sector(HS4)-year fixed effects, which absorb the level of UIP deviations and every other market-level determinant of switching. The sample covers 2008–2013. Standard errors in parentheses are clustered at the firm-product-origin level, except Column (3), clustered at the firm level. The hypothesis of equal CIA and LC interactions is rejected with $p = 0.0001$ in Column (5) and $p = 0.0029$ in Column (7). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

both samples). The negative level coefficients carry the complementary message: outside the crisis window, trade-finance relationships switch less – dollar financing arrangements anchor invoicing to the dollar until the funding advantage that sustains them reverses.

5.6 Additional Results and Discussion

Exchange Rate Risk Hedging. Table 17 in Appendix A.11 re-estimates Equation (13) replacing UIP deviations with CIP deviations, computed using one-year-ahead forward exchange rates between US dollar and Peso. This exercise addresses the concern that firms hedge through forward markets and it separates the two mechanisms of the model. In Equation (11), firm exposure enters only if $\Sigma > 0$, so the operational-hedging margin operates entirely through exchange rate risk. Strategic complementarities instead operate through the fixed-cost differential and carry no exchange rate risk. Because CIP deviations are constructed off the forward rate, they isolate the funding wedge net of exchange rate risk and therefore shut down the first channel while leaving the second intact. The estimates match this prediction: the interaction with strategic complementarities remains negative and significant throughout, while the interaction with natural hedging is insignificant or has a positive sign.

Bargaining in Invoicing. The invoicing pattern observed in the data implies that exporters located in the origin countries started to accept payments in Chilean peso. Arguably, Chilean pesos are a less common currency compared to the ubiquitous US dollar. This suggests the existence of a relatively higher bargain power of Chilean importers when defining which currency to use to exchange goods internationally. We provide evidence that is coherent with the existence of a higher relative bargain power of Chilean importers with respect to foreign exporters.

First, the left panel of Figure 15 in Appendix A.5 shows a positive relationship between the cumulative change in the aggregate share of Chilean peso at the origin level and the Chilean export share in the origin's total exports in 2008. Intuitively, the graph suggests that countries whose total exports were more dependent on Chile also tended to have a higher share of transactions invoiced in Chilean currency. This aligns with the idea that Chile was able to exert its currency influence over trading partners that were more reliant on it.

Second, a model with strategic interactions in pricing among Chilean importers predicts an inverted U-shape relationship between exchange rate pass-through to import prices invoiced in local currency and importer's market share (Juarez, 2024). We confirm the existence of this pattern in the right panel of Figure 15 in Appendix A.5.

CLP-Euro Invoicing. The theoretical mechanism connecting funding costs to invoicing decisions in Section 4 is not specific to the dollar and applies to any currency pair. The euro provides a within-sample test of this mechanism in our data. We re-estimate our main specification with the euro in the dollar's slot: the dependent variable is an indicator equal to one for Chilean peso invoicing and zero for euro invoicing; the funding-cost wedge is the realized euro–peso UIP deviation constructed from the 3-month Euribor; and the strategic complementarity and natural hedge measures are rebuilt using euro-invoicing competitors and euro-invoiced exports, respectively, while leaving everything else, including the instruments, unchanged.[28]

Table 18 in the Appendix A.12 reports the results. Every coefficient has the same sign, and is close in magnitude, to its dollar counterpart: invoicing shifts toward the peso when euro funding is relatively expensive; euro-coordinated markets retain the euro, with this retention weakening as the funding-cost wedge rises; and euro receivables from the origin dampen the response, just as dollar receivables do in our baseline. The broad consistency with the

[28]Because Chilean exports are dollar-invoiced almost everywhere but euro-invoiced essentially only on European routes, we measure the euro hedge at the firm–origin level.

dollar-based results indicates that the funding-cost, coordination, and hedging mechanisms highlighted in the theoretical framework are properties of currency invoicing more generally, rather than of the dollar–peso pair specifically.[29]

Evidence Across Countries. We construct realized UIP deviations against the US dollar as in Equation (12) for 113 countries, using interest rates and bilateral exchange rates from IMF International Financial Statistics and US benchmark rates from FRED. Matching these series to the invoicing data leaves 56 countries in the estimation sample.[30] Figure 21 shows the same pattern we document for Chile: the median deviation fluctuates around -5 percentage points before the crisis, turns positive in early 2008 and peaks at about 12 percentage points, before reverting by 2009.

We combine these series with country-level invoicing data from Boz et al. (2025) and estimate the cross-country counterpart of Equation (13), in which the firm-level determinants have no counterpart and are omitted. Figure 22 and Table 19 in Appendix A.13 report the results. The coefficient on UIP deviations is positive throughout and rises as fixed effects absorb more confounding variation, reaching 1.80 with country and year fixed effects and is essentially unchanged when the sample is restricted to emerging markets. It is not statistically significant at conventional level, which is not surprising given that the invoicing data are sparse and unbalanced. However, taken together, we read this evidence as suggestive corroboration of the mechanism operating beyond Chile, rather than as independent identification of it.

6 Firm-level Effects of Switching and Aggregate Implications

Real Effects of Switching. We assess whether the switch to peso invoicing is associated to different trade outcomes at the importer level or only the currency of invoicing. We compare firms that adopted the peso in the period 2008–2012 after importing exclusively in US dollars with importers that never invoice in pesos. The top panels in Figure 23 in Appendix A.14 reports the estimates at the firm-origin-product level. Within continuing importers and conditional on origin-sector-year fixed effects, quantities and unit values display no differences before peso adoption, and remain statistically indistinguishable also in the post-adoption window. We also find no significant effects on price or quantity dynamics in the post-adoption period when comparing imports denominated in Chilean pesos with the

[29] Empirically, we notice that this exercise represents a replication on held-out data rather than as an identification exercise for a euro-specific shock, since the euro–peso wedge co-moves with the dollar–peso wedge over this period.

[30] We follow the proxy hierarchy of Kalemli-Ozcan and Varela (2021), using deposit rates where their coverage exceeds five years, then money market rates, then treasury bill rates. We exclude currency unions and aggregates, euro-denominated economies, and economies operating a long-standing hard peg. The construction and the sign convention are identical to those used for Chile.

same firms' imports that remain invoiced in dollars.[31] The invoicing switch that followed the GFC was thus a market-driven change in the invoicing currency, leaving prices and quantities largely unchanged.

Price Elasticities to Exchange Rate. Here, we quantitatively assess the macroeconomic implications of the large changes in invoicing shares after the GFC. We focus on the implications of invoicing choices for the dynamics of the terms of trade and import inflation. We proceed as follow. We first estimate short-run and long-run price elasticities to exchange rate fluctuations accounting for the different invoicing currencies, building on existing empirical frameworks (Adler et al., 2020; Chen et al., 2022; De Gregorio et al., 2024). We then leverage the estimated elasticities and the evolution of invoicing shares to conduct accounting exercises (Auer et al., 2021).[32] We find that the change in the invoicing shares after the GFC reduced the sensitivity of import inflation to bilateral nominal exchange rate fluctuations by 5 percent, while increasing that of the terms of trade.

We first estimate the exchange rate pass-through to border prices, both in the short-run and in the long-run, conditioning on the currency of invoicing. Following existing econometric frameworks, the benchmark specification is:

$$\Delta P^{k}_{fopt} = \sum_{l=0}^{8} \beta_l^{LCP} \Delta e_{t-l}^{CLP/o} \times D^{LCP}_{fopt} + \sum_{l=0}^{8} \beta_l^{PCP} \Delta e_{t-l}^{CLP/o} \times D^{PCP}_{fopt} + \\ + \sum_{l=0}^{8} \beta_l^{VCP} \Delta e_{t-l}^{CLP/\$} \times D^{VCP}_{fopt} + FE_{fopt} + \varepsilon_{fopt}, \tag{22}$$

where $\Delta x_t \equiv x_t - x_{t-1}$. Thus, ΔP^k_{fopt} denotes the log change in unit values (expressed in Chilean pesos) of product p imported by firm f from origin o between quarters $t-1$ and t, invoiced in currency k. An increase in $\Delta e_t^{CLP/o}$ corresponds to a depreciation of the Chilean peso relative to the currency of country o. The dummy variables D^{LCP}, D^{PCP}, and D^{VCP} indicate whether the transaction is invoiced in local, producer, or vehicle currency, respectively. For a given invoicing scheme, short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^{8} \beta_l$. We re-estimate Equation (22) using export unit values as dependent variable.

Table 7 reports the short-run and long-run exchange rate pass-through rates into import and export prices estimated from the main specification in Equation (22). Pass-through into import prices is not statistically different from zero in the short run when transactions are invoiced in local currency, and rises to roughly 0.56 in the long run; pass-through is instead almost complete when transactions are invoiced in vehicle and partner currencies in the short

[31] In the bottom-left panel of Figure 23, import quantities for firms that later switch to peso invoicing steadily converge toward those of the same firms' dollar-invoiced imports before adoption, predating the currency switch and reflecting the selection of expanding relationships into peso invoicing.

[32] The accounting approach is consistent with the firm-level evidence on price and quantities: the aggregate consequences of the invoicing decisions operate through the pricing regime attached to each currency, not through changes in trade itself.

run, but declines after two years. On the export side, pass-through rates are almost complete in the short-run when transactions are invoiced in dominant and local currencies. We do not report the pass-through rate into export prices conditional on being invoiced in producer currency (i.e. Chilean peso) because of a very restricted sample. In the long-run, the pass-through rate decreases to approximately 50-70 percent in both cases. These patterns are consistent with the idea that prices are sticky in the currency in which they are invoiced (Chen et al., 2022; Gopinath et al., 2010; De Gregorio et al., 2024).

Table 7: Price Sensitivities to Exchange Rates

	Imports		Exports	
Invoicing	Short-run	Long-run	Short-run	Long-run
LCP	0.062 (0.068)	0.559 (0.260)	0.879 (0.029)	0.675 (0.088)
PCP	0.884 (0.016)	0.703 (0.055)	. (.)	. (.)
VCP	0.879 (0.014)	0.389 (0.037)	0.869 (0.023)	0.503 (0.045)

Note: The left (right) panel show the estimated coefficients from the specification in Equation (22) using import (export) prices as dependent variable. We use data from official Chilean customs running from 2007 to 2019 for imports and 2009 to 2019 for exports. We exclude the PCP case for export prices because of the absence of sufficient observations in the estimation. Values in parenthesis represents the standard errors clustered at the origin-time level. Short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^{8} \beta_l$.

Effects on Terms of Trade and Import Inflation. In standard models of international economics (e.g. Mundell-Fleming), in which international prices are invoiced in the exporter's currency, exchange rates play a key role in external adjustment. However, the widespread use of US dollar invoicing, together with the differences in price elasticities documented above, impacts how the terms of trade and import inflation respond to exchange rate fluctuations.

Using accounting exercises as in Auer et al. (2021), we define the sensitivity of terms of trade at horizon l to a depreciation of the Chilean peso in terms of invoicing shares and price elasticities as follows:

$$\begin{aligned}\Delta TOT_t^l =& \Delta e \left[\sum_{\tau=0}^{l} \beta_\tau^{PX,CLP} \times S^{X,CLP} + \beta_\tau^{PX,D} \times S^{X,D} + \beta_\tau^{PX,p} \times S^{X,p}\right] - \\ & \Delta e \left[\sum_{\tau=0}^{l} \beta_\tau^{PM,CLP} \times S^{M,CLP} + \beta_\tau^{PM,D} \times S^{M,D} + \beta_\tau^{PM,p} \times S^{M,p}\right],\end{aligned} \tag{23}$$

where $S^{X,z}$ ($S^{M,z}$) the share of exports (imports) invoiced in currency z, and $\beta_l^{j,z}$ is the elasticity of j to exchange rates when invoiced in currency z at horizon l, with j being import and export prices ($\{PX, PM\}$) and $z \in \{CLP, p, D\}$. Similarly, we can define the response of the import price index (IPI) to a depreciation of the Chilean peso. For each horizon l, we use the following formula:

$$\Delta IPI_t^l = \Delta e \left[\sum_{\tau=0}^{l} \beta_\tau^{PM,CLP} \times S^{M,CLP} + \beta_\tau^{PM,D} \times S^{M,D} + \beta_\tau^{PM,p} \times S^{M,p}\right]. \tag{24}$$

Table 8: ToT and IPI Sensitivity to Exchange Rates

	Import prices		Terms of trade	
Year	Short-run	Long-run	Short-run	Long-run
2007	0.876	0.464	-0.006	0.065
2019	0.832	0.457	0.033	0.073

Notes: The left (right) panel reports the response of the import price index (terms of trade) to a depreciation of the Chilean peso using Equation (24) (Equation (23)). We use the estimated coefficients from Equation (22) and reported in Table 7. Price elasticities not statistically different from zero are set to zero. Short-run ERPT coefficients are equal to β_0 while the long-run ERPT is the 8-period cumulative sum, $\sum_{l=0}^{8} \beta_l$. Invoicing shares are computed from official import (export) Chilean customs data from 2007 (2009) for the initial period and from 2019 for the last period.

Table 8 shows that, relative to 2007, the import price index of Chile has become less sensitive to exchange rate fluctuations while the sensitivity of terms of trade has increased in 2019. The IPI sensitivity has decreased by approximately 5 percentage points in the short-run and 1.5 percentage points in the long-run. The sensitivity of the terms of trade to a one percent depreciation of the Chilean peso increased to 0.03 pp from zero in the short-run, and by 11 percent in the long run (0.073 vs 0.065). The key driver of these changes is the rise in the share of imports invoiced in local currency between 2007 and 2019 (Figure 1), which carries a lower pass-through to exchange rate fluctuations. Thus, abstracting from invoicing choices improperly informs on the effects of exchange rate fluctuations on key macroeconomic variables (Gopinath and Itskhoki, 2022; Barbiero, 2021).

Conclusion

Our analysis reveals how temporary shocks to dollar financing availability can weaken the US dollar invoicing incentives in international trade and boost alternative forms of invoicing, challenging the dominance of the US dollar. We document a gradual decline in the share of Chilean imports invoiced in US dollars over a decade after the spike in the cost of dollar financing during the Global Financial Crisis, accompanied by a corresponding increase in Chilean peso invoicing. Decomposition exercises show that surviving firms contribute approximately 80 percent of the observed shift. Our model explains firms' invoicing decisions based on the relative costs of financing in domestic versus vehicle currencies, which can be proxied by UIP deviations, treating imported inputs as working capital. We show that temporary shocks to UIP deviations influence invoicing patterns, with sunk costs introducing hysteresis, permanently altering the convenience of using different currencies. We test the key mechanisms of our theory using firm-level data and various econometric techniques. Our findings support the key mechanisms, explaining half of the observed invoicing patterns in Chilean imports.

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A Appendix: Data & Robustness

A.1 Summary Statistics

Table 9: Number of Invoicing Currencies within Firm-Product-Origin-Year Quartet

No. of Currencies	*fpot* Quartets	Share of Transactions (%)	Share of Value (%)
1	8384867	96.2	92.4
2	318226	3.65	7.07
3	16185	0.19	0.45
4	228	0.0026	0.029

Notes: The Table reports summary statistics on multi-currency use at the firm-product-origin-year ($fopt$) level. Specifically, we count the total number of different currencies that a firm uses to import a product (defined at the HS8 level) from a given origin in a given year. Data from Chilean Customs Agency from 2007 to 2019.

Table 10: Number of Invoicing Currencies within Firm-Product-Destination-Year Quartet

No. of Currencies	*fpot* Quartets	Share of Transactions (%)	Share of Value (%)
1	1225319	97.3	95.2
2	32726	2.60	4.73
3	1116	0.089	0.12

Notes: The Table reports summary statistics on multi-currency use at the firmproductoriginyear ($fopt$) level. Specifically, we count the total number of different currencies that a firm uses to export a product (defined at the HS8 level) from a given destination in a given year. Data from Chilean Customs Agency from 2009 to 2019.

Table 11: Summary Statistics for USD-CLP FX Forecasts

	StDev	Min	Max	N. Forecasts
2006	22.14	525	620	64
2007	27.97	480	637	119
2008	56.07	408	710	120
2009	59.01	458	710	140
2010	34.12	440	620	135
2011	24.24	425	559	154
2012	19.53	450	576	200
2013	24.37	450	550	184
Observations	1116			

Notes: The Table reports summary statistics of the one-year expected USD-CLP exchange rate. For each year we report the number of forecasters, the range of their forecast and the standard deviation of their forecasts. Data are sourced from Bloomberg.

A.2 Additional Decompositions

Table 12: Dynamic Olley-Pakes Decomposition with Entry and Exit: Imports at Origin Level

	USD	Peso	Euro	Others
– Aggregate Invoicing Share	-8.397	5.114	2.919	0.364
Contribution Net Entry				
Total	-1.794	0.822	0.808	0.164
Firm Entry	-2.644	0.901	1.867	-0.124
Firm Exit	0.850	-0.078	-1.059	0.288
Contribution Incumbents				
Total	-6.603	4.292	2.111	0.200
Within Firm	-4.403	5.151	-0.780	0.032
Between Firm	-2.200	-0.859	2.891	0.168
Within Firm				
Total	-4.403	5.151	-0.780	0.032
Net Entry Origin	-0.581	0.170	0.491	-0.080
Within Origin	-4.536	4.294	0.105	0.137
Between Origin	0.714	0.687	-1.377	-0.024

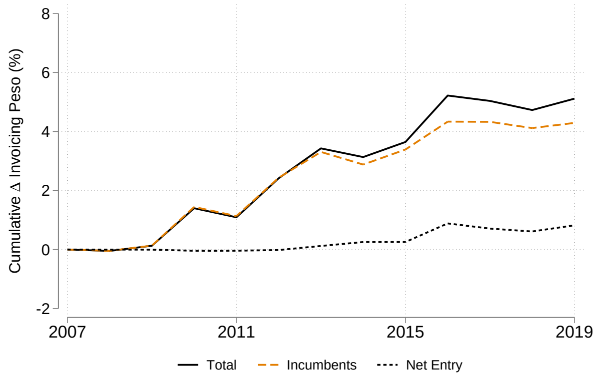
Notes: Table 12 reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the origin level for Chilean imports between 2007 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3) where Equation (3) is properly adjust to consider only the origin level. One time period corresponds to one year.

Table 13: Olley-Pakes Decomposition of Exports Invoicing

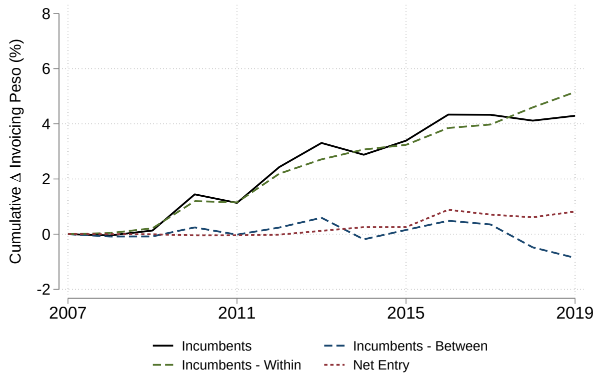
	USD	Peso	Euro	Others
Δ Aggregate Invoicing Share	-1.329	0.010	0.466	0.852
Net Entry	0.344	-0.006	-0.055	-0.283
Firm Entry	0.528	0.017	-0.506	-0.039
Firm Exit	-0.184	-0.023	0.451	-0.244
Contribution Incumbents	-1.673	0.017	0.521	1.135
Within Firm	3.540	-0.237	-4.415	1.112
Between Firm	-5.213	0.254	4.936	0.023
Within Firm				
Net Entry Destination-product	-2.603	2.205	0.230	0.168
Within Destination-product	2.581	-2.426	-1.832	1.677
Between Destination-product	3.561	-0.016	-2.813	-0.733

Notes: The Table reports the results from the dynamic Olley-Pakes decomposition with entry and exit at the destination level for Chilean exports between 2009 and 2019. It corresponds to the decompositions in Equations (1), (2), and (3). Products are defined at the HS8 level, the most granular classification available in the dataset. We report the cumulative decomposition. Numbers are in percentage. Data are from the Chilean Customs Agency.

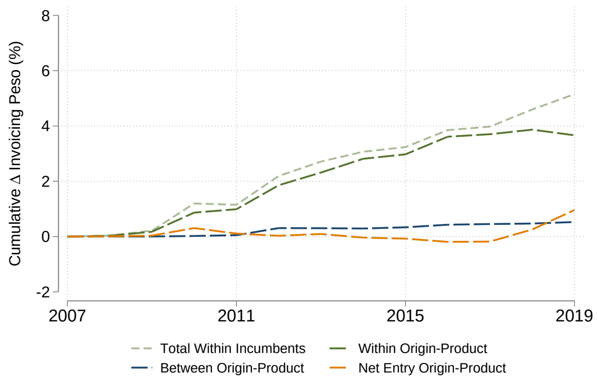
Figure 7: Decomposing Chilean Peso Invoicing Share over Time



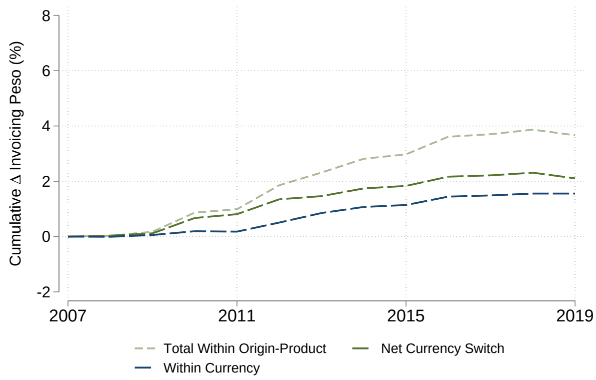
Panel (A): First Layer



Panel (B): First Layer



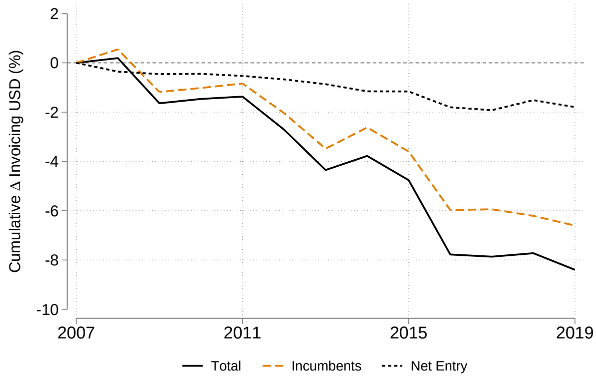
Panel (C): Second Layer



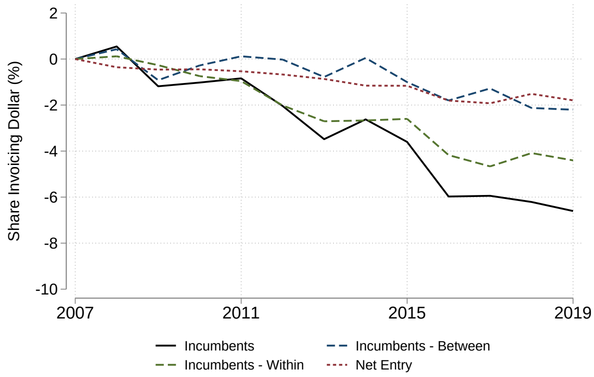
Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic OlleyPakes decomposition from Equations (1), (2), and (3) at the origin-product level for Chilean imports between 2007 and 2019. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in Chilean peso. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year. Data are from the Chilean Customs Agency.

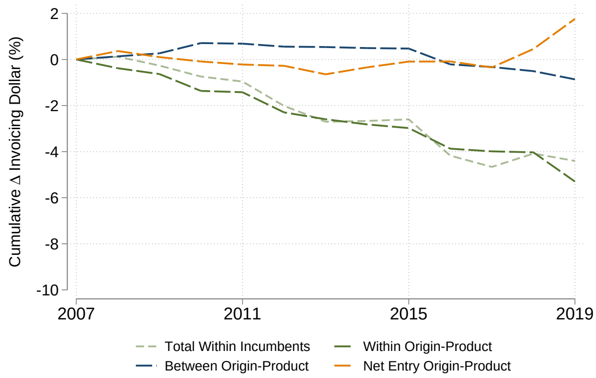
Figure 8: Decomposing US Dollar Invoicing Share over Time



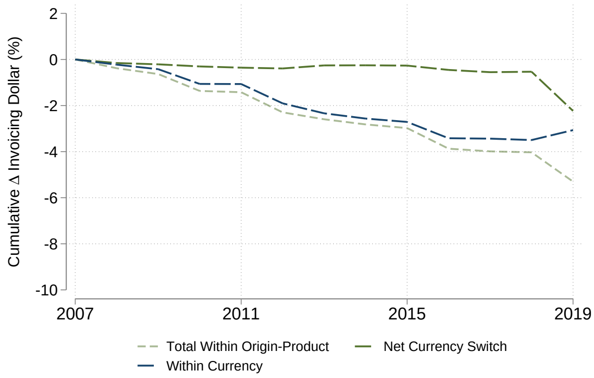
Panel (A): First Layer



Panel (B): First Layer



Panel (C): Second Layer

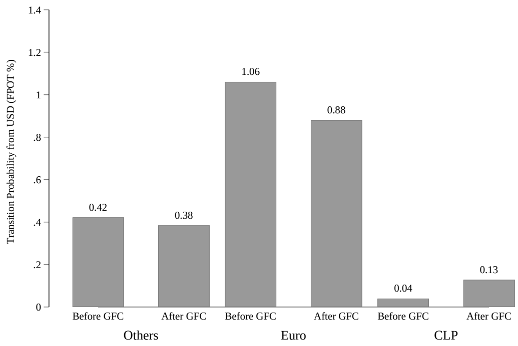


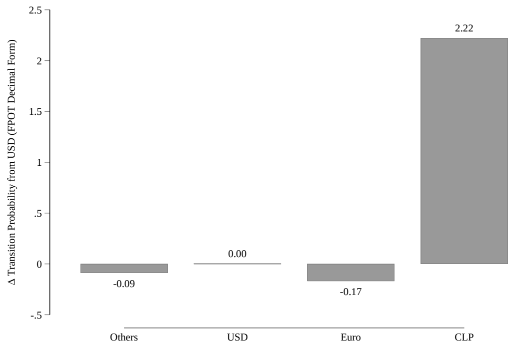
Panel (D): Third Layer

Notes: The Figure reports the results from the dynamic Olley-Pakes decomposition from Equations (1), (2), and (3) at the originproduct level for Chilean imports between 2007 and 2019. Each panel corresponds to a layer of the decomposition. We consider the share of imports invoiced in US dollar. Products are defined at the HS8 level, the most granular classification available in the dataset. One time period corresponds to one year. Data are from the Chilean Customs Agency.

A.3 Additional Results on Invoicing and Switching Probability

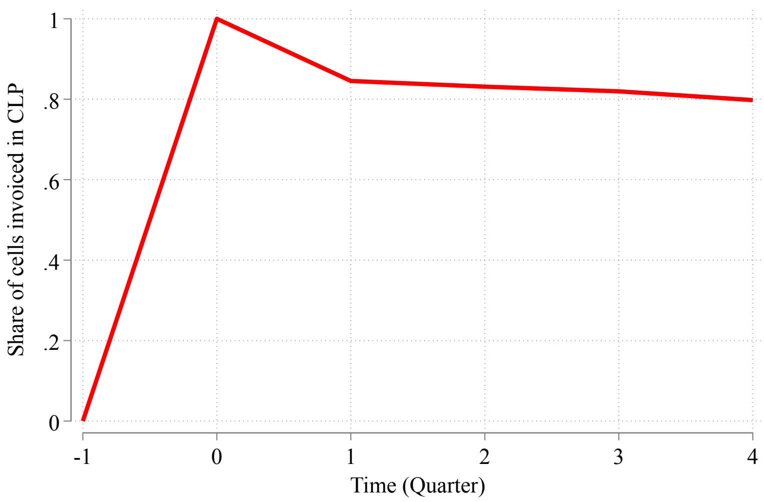
Figure 9: Transition Probability from USD to CLP





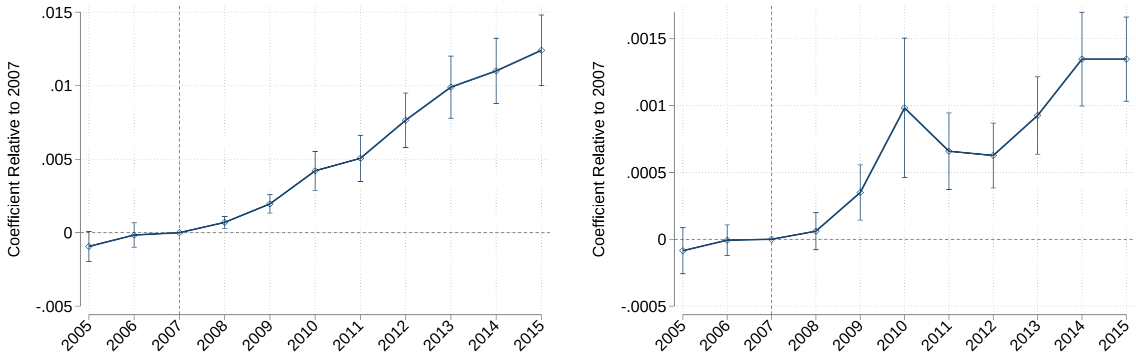
Note: The Figure reports transition probabilities across invoicing currencies for firm–origin–product cells in the annual customs data. Each cell-year observation is paired with the cell's next observed year, and a transition is assigned to a period when both years fall within it: before the GFC (transitions from 2004 to 2008) and after the GFC (from 2008 to 2012). The left panel reports the average transition probability from US dollar invoicing to each alternative currency group, in percent, where "Others" collects all currencies other than the US dollar, the Euro and the Chilean peso. The right panel reports the change in these transition probabilities between the two periods, expressed as a proportion of the pre-GFC level: the value of 2.22 indicates that the transition probability from the US dollar to the Chilean peso more than tripled, while it declined toward the Euro and all other currencies. Cells invoiced in more than one currency within a given year are excluded. Data are from the Chilean Customs Agency.

Figure 10: Persistence in Switching from US dollar to Chilean peso



Note: The Figure describes the invoicing currency in importerorigin-product im cells that switch for the first time from US dollars to Chilean peso in their imports to the four quarters after the switch. For each im cell, quarter zero corresponds to the first quarter in which the invoice currency is Chilean peso. We then compute the share of im cells invoicing in Chilean peso in each period. Data are from the Chilean Customs Agency.

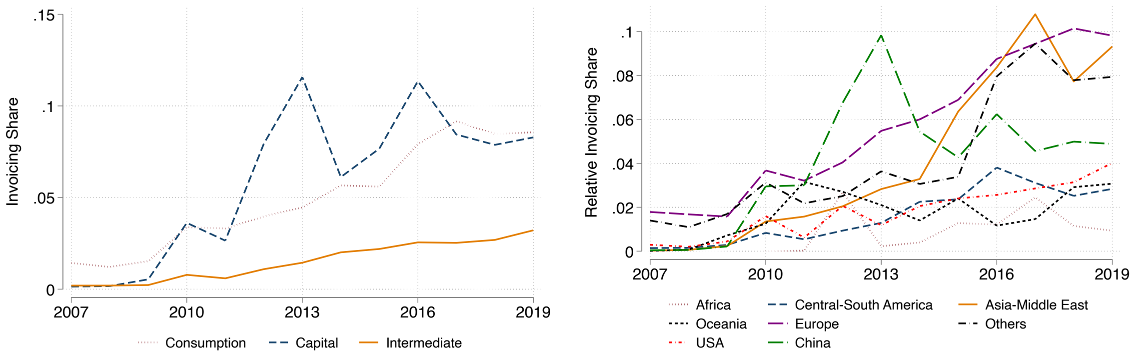
Figure 11: Effect of GFC on Probability of Invoicing in Chilean peso: Annual Frequency



Note: The Figure plots the estimated coefficients from the annual analogue of Equation (4), estimated over 2005–2015 with year fixed effects in place of the quarter-year fixed effects. Each point represents the effect of a given year on the probability of invoicing in Chilean peso, relative to 2007, the last year before the GFC. The left panel reports results for the full sample, while the right panel restricts the sample to single-currency importer–market pairs invoiced in US dollars in the previous year, so that each coefficient measures the change in the probability of switching from US dollar to Chilean peso invoicing. The vertical dashed line marks the 2007 reference year. 95 percent confidence intervals with standard errors clustered at the importer–origin level are reported. Data are from the Chilean Customs Agency, as compiled by Garcia-Marin et al. (2019).

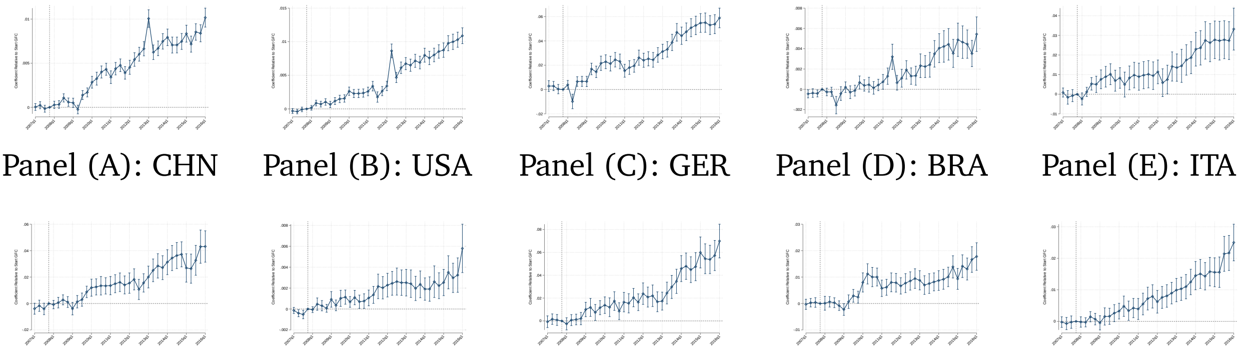
A.4 Additional Results: Product and Origin

Figure 12: Subgroup by Origin or Product Type - Chilean Peso Invoicing



Notes: The left panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within three broad product categories: Consumption, capital, and investment goods as defined according to the BEC classification. The right panel of the Figure shows the evolution of the invoicing share in Chilean Peso in imported goods within major trade partners: USA, Europe, China, Central-South America, Asia-Middle East, Africa and Others. Data are from the Chilean Customs Agency.

Figure 13: Effect of GFC on Probability of Invoicing in Chilean Peso: Main Origins

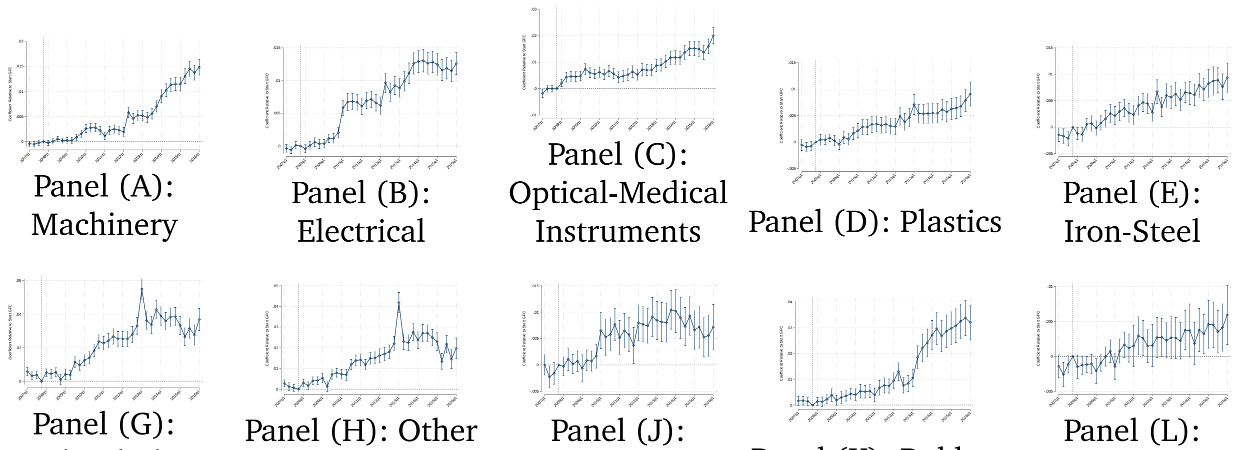


Panel (A): CHN | Panel (B): USA | Panel (C): GER | Panel (D): BRA | Panel (E): ITA

Panel (F): ESP | Panel (G): ARG | Panel (H): FRA | Panel (I): JAP | Panel (J): MEX

Notes: The Figure reports the results from Equation (4) estimated separately for each of top ten origin countries in the sample (ranked by number of observations at the $fopt$ level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2007Q4. Standard errors are clustered at the importerproductorigin level. Data are from the Chilean Customs Agency.

Figure 14: Effect of GFC on Probability of Invoicing in Chilean Peso: Main HS2



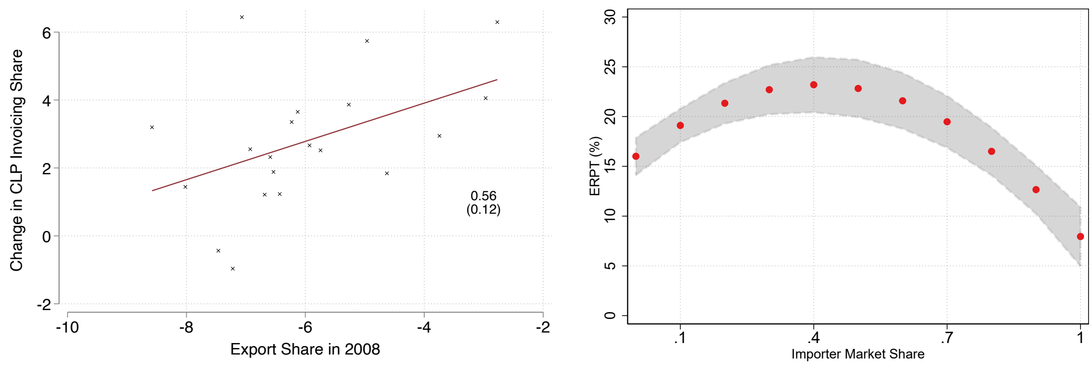
Panel (A): Machinery | Panel (B): Electrical | Panel (C): Optical-Medical Instruments | Panel (D): Plastics | Panel (E): Iron-Steel

Panel (G): Textile-Clothing | Panel (H): Other fabrics | Panel (J): Furniture | Panel (K): Rubber | Panel (L): Furniture

Notes: The Figure reports the results from Equation (4) estimated separately for each of top ten HS2 products in the sample (ranked by number of observations at the $fopt$ level between 2007Q1 and 2019Q4). The dependent variable is the probability of invoicing in Chilean Peso, relative to 2007Q4. Standard errors are clustered at the importerproduct level. Data are from the Chilean Customs Agency.

A.5 Additional Results: Importers' Bargaining Power in Invoicing

Figure 15: Invoicing as Importers' Choice



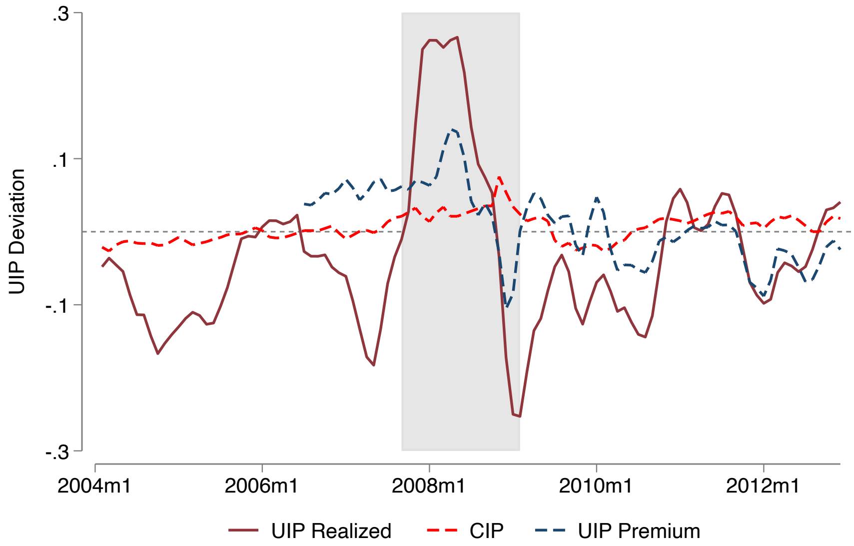
Notes: The left panel plots the relationship between the cumulative change in the aggregate share of CLP at the origin level and the Chilean export share in the origin's total exports in 2008 (in logs). We compute the cumulative change starting from 2008. We consider all horizons from 5 to 10 years, and plot the relationship after absorbing time fixed effects. The right panel plots the estimated quadratic relationship between the importer's market share and the pass-through rate of exchange rate fluctuations in import prices, estimated using the following specification:

$$\Delta \log p_{foit} = \beta_1 \Delta \log e_{ot} + \beta_2 S_{fsit} + \beta_3 \Delta \log e_{ot} \times S_{fsit} + \epsilon_{foit},$$

where S_{fst} is the market share of importer f in sector s (HS4-origin), and $\log p_{foit}$ is the unit import price in local currency for importer f, product i (HS8) from origin o.

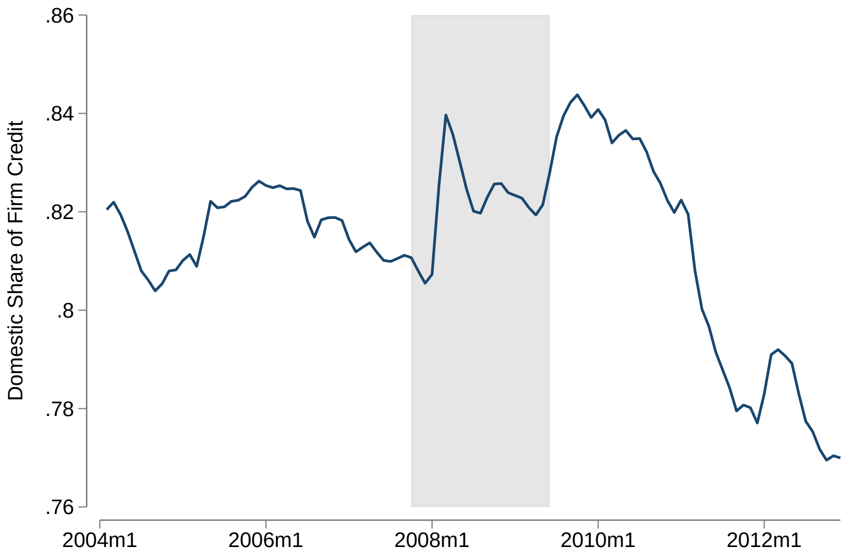
A.6 Additional Results: Model Validation

Figure 16: UIP Between the US dollar and Chilean peso



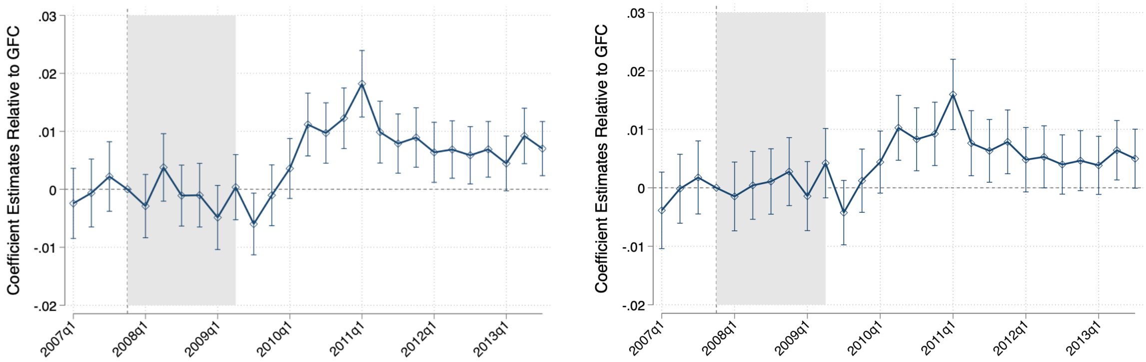
Note: The Figure plots the one-year ahead UIP deviation (dash blue), the realized one-year ahead UIP deviation (solid red), and the one-year CIP (dashed red) between the Chilean peso and the US dollar. A positive value indicates that borrowing in Chilean pesos is relatively cheaper than borrowing in US dollars. Interest rates are measured using commercial lending rates on short-term (less than one year) loans denominated in US dollars and Chilean pesos available from domestic banks. We construct the one-year-ahead expected exchange rate averaging the professional forecasters' expectations at the monthly frequency. We use non-deliverable forward (NDF) outright contracts for the U.S. dollar-peso from Bloomberg to construct measures of covered interest parity. Exchange rate forecasts are obtained from Bloomberg, available from 2006 onward. Table 11 reports summary statistics. Data are at the monthly frequency and smoothed using a 2-month rolling window.

Figure 17: Domestic-Currency Share of Bank Credit to Firms



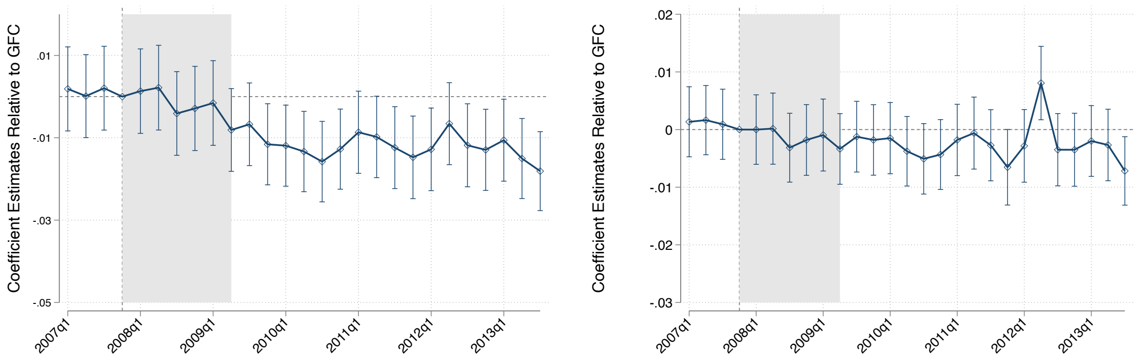
Note: The Figure plots the share of domestic bank credit to firms denominated in domestic currency, at monthly frequency. Credit to firms is defined as commercial loans plus foreign-trade loans. The foreign-currency value is published converted into pesos and enters the denominator of this share, so we value it at a fixed exchange rate, so that the share moves with quantities of credit rather than with the exchange rate. Data are smoothed using a two-month rolling window, as in Figure 5. The grey shaded area covers 2007Q4–2009Q2. Data are from the *Comisión para el Mercado Financiero*, as published in the Banco Central de Chile statistical database.

Figure 18: Event Study - Dollar Financing Need



Note: The Figure plots the coefficients β_s from the event study specification in Equation (17), capturing the differential increase in the probability of Chilean peso invoicing due to the firm's dollar financing need, measured by the share of imports denominated in US dollar over the firm's active quarters between 2007Q1 and 2007Q3 (Proposition 2). Both panels replace the time fixed effects of the benchmark specification with more demanding structures. The left panel includes origin-time fixed effects, which absorb shocks common to all Chilean imports from a given origin. The right panel includes origin-product-time fixed effects, so that identification comes from comparing firms sourcing the same product from the same origin in the same quarter and differing only in their pre-crisis dollar exposure. All specifications include firm-product-origin fixed effects. Coefficients are normalized to zero in 2007Q4, marked by the vertical dashed line; the shaded area covers 2007Q4–2009Q2. We report 95 percent confidence intervals. Standard errors are clustered at firm-product-origin level.

Figure 19: Event Study - Strategic Complementarity



Note: The Figure plots the coefficients β_s from the event study specification in Equation (17), using the leave-one-out strategic complementarity measure of Equation (14), averaged over the firm-market's active quarters between 2007Q1 and 2007Q3 (Proposition 3). Both panels replace the time fixed effects of the benchmark specification with more demanding structures. The left panel includes sector(HS4)-time fixed effects, which absorb shocks common to all firms in a sector. The right panel includes firm-time fixed effects, which absorb every firm-level shock and identify the coefficients from within-firm comparisons across markets differing in the dollar intensity of competitors. All specifications include firm-product-origin fixed effects. Coefficients are normalized to zero in 2007Q4, marked by the vertical dashed line; the shaded area covers 2007Q4–2009Q2. We report 95 percent confidence intervals. Standard errors are clustered at firm-product-origin level.

A.7 Additional Results: UIP and Pre-GFC Placebo

Table 14: Pre-GFC Placebo: UIP and Peso Invoicing

	Full Sample (2004–2014)			Pre-GFC Placebo (2004–2007)		
	(1)	(2)	(3)	(4)	(5)	(6)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
$UIP_t^{\$,PESO}$	0.027***	0.026***	0.026***	0.023***	0.022***	0.021***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
$NH_{ft}^{\$}$		-0.012***	-0.012***		-0.009***	-0.009***
		(0.001)	(0.001)		(0.001)	(0.001)
$SC_{kt}^{\$}$			-0.005***			0.001
			(0.001)			(0.001)
Adjusted R^2	0.419	0.419	0.417	0.437	0.438	0.439
Firm-Origin-HS8	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1075005	1075005	1054967	327439	327439	320459

Notes: OLS regressions. The dependent variable is a dummy variable equal to 1 when the $fopt$ transaction is invoiced in Chilean peso and 0 in U.S. dollar. $UIP_t^{\$,PESO}$ denotes deviations from UIP as defined in Equation (12). $SC_{kt}^{\$}$ is the strategic complementarity index as in Equation (14) and $NH_{ft}^{\$}$ measures the natural hedge of the firm as the firm-level share of exports invoiced in U.S. dollars. Sample and variable construction are identical to Table 2; Columns (2)–(3) and (5)–(6) additionally control for importer size (firm f's share of aggregate imports). Firm-Origin-HS8 fixed effects are included in all specifications. Standard errors are clustered at the firm-product-origin level. Columns (1)–(3): full sample (2004–2013, the period for which the UIP measure is available). Columns (4)–(6): pre-GFC sub-sample (2004–2007). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.8 Additional Results: Sunk Cost

Table 15: Sunk Cost Persistence After UIP Normalization

	(1)	(2)	(3)	(4)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
Lagged Peso Share	0.402***	0.399***	0.423***	0.418***
	(0.006)	(0.007)	(0.013)	(0.013)
Lagged Peso Share x Post-2010			-0.021*	-0.020
			(0.012)	(0.012)
Adjusted R^2	0.537	0.533	0.537	0.533
Controls	No	Yes	No	Yes
Firm-Origin-HS8	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Observations	2923670	2883055	2923670	2883055

Note: The dependent variable equals 1 if the $fopt$ transaction is invoiced in Chilean pesos and 0 otherwise. "Lagged Peso Share" refers to the one-year lagged share of imports invoiced in local currency at the firm level, as in Equation (18). The table reports coefficients from the specification in Equation (18), augmented with an interaction term between the Lagged Peso Share and a Post-2010 dummy, equal to 1 from 2010 onward and 0 otherwise. Sample and variable construction are identical to Table 3. Controls include: (i) a natural hedging measure, defined as the share of exports invoiced in US dollars; (ii) a strategic complementarity index, defined as the average share of imports invoiced in US dollars among all competitors in a given sector(HS4)-origin pair, as in Equation (14); and (iii) a dummy equal to 1 if the importer also exports and 0 otherwise. The sample covers 2004–2015 and is from Garcia-Marin et al. (2019). Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.9 Additional Results: Probability of Switching

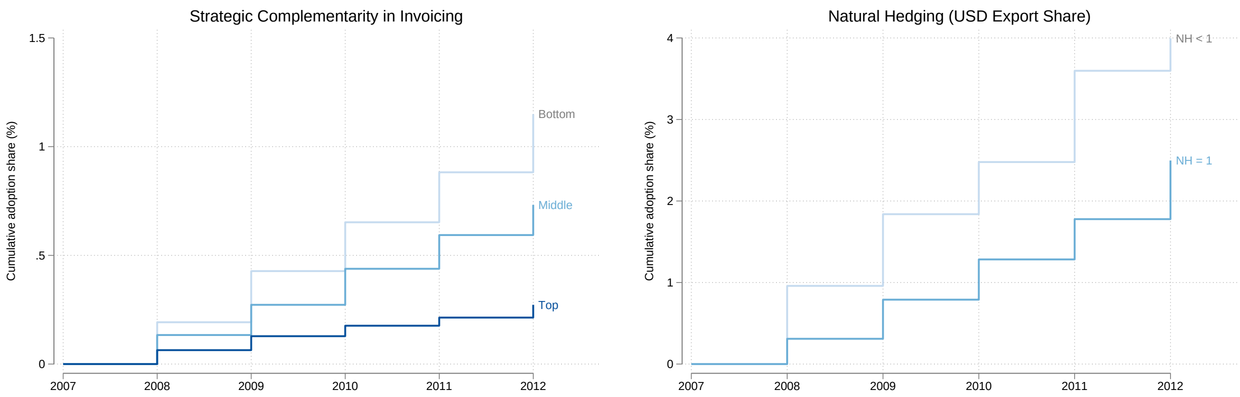
Table 16: Switching: Cross-Sectional Determinants

	OLS		IV	
	(1)	(2)	(3)	(4)
	$Switch_{fopt}$	$Switch_{fopt}$	$Switch_{fopt}$	$Switch_{fopt}$
$UIP_t^{\$,PESO}$	0.0186***	0.0569***	0.2571***	
	(0.0014)	(0.0133)	(0.0813)	
$SC_{kt}^{\$}$		-0.0056***	-0.0591***	-0.0464***
		(0.0011)	(0.0143)	(0.0146)
$NH_{ft}^{\$}$		-0.0072***	-0.0068***	-0.0066***
		(0.0009)	(0.0009)	(0.0009)
$SC_{kt}^{\$} \times UIP_t^{\$,PESO}$		-0.0063	-0.2452**	-0.2067**
		(0.0071)	(0.0968)	(0.0987)
$NH_{ft}^{\$} \times UIP_t^{\$,PESO}$		-0.0357***	-0.0243*	-0.0246*
		(0.0116)	(0.0133)	(0.0132)
Firm-Origin-HS8	Yes	Yes	Yes	Yes
Year	No	No	No	No
Weak IV F-Stat			182.1876	174.6385
Observations	597827	588966	588966	588966

Note: The Table re-estimates Equation (13) replacing the invoicing indicator with the switching outcome. The dependent variable equals 1 when a firm-product-origin combination invoiced imports exclusively in US dollars at $t-1$ and invoices in Chilean peso at t, on the set of combinations at risk of switching (mean 0.0033). Columns (1) and (2) report OLS estimates. Columns (3) and (4) instrument $SC_{kt}^{\$}$ and its interaction with $UIP_t^{\$,PESO}$ using the leave-out shifter of Equation (15) and its UIP interaction. Column (4) additionally includes year fixed effects, which absorb the level of $UIP_t^{\$,PESO}$. All regressors, the sample restriction to exporting firms, the firm size control (included, not reported), and the firm-origin-product fixed effects are as in Table 2. The sample covers 2004–2014 from Garcia-Marin et al. (2019). Standard errors are clustered at the firm-product-origin level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.10 Additional Results: Adoption Ordering by Pre-Crisis Determinants

Figure 20: Cumulative Adoption of Peso Invoicing by Pre-Crisis Determinants



Note: The Figure plots the cumulative share of firms that have adopted Chilean peso invoicing in imports by each year, by pre-crisis exposure group. The sample consists of firms that invoice an import in Chilean peso for the first time between 2008 and 2012 and of never-adopting firms with US dollar import activity; firms with peso use before 2008 are excluded by construction. The left panel groups firms by the strategic complementarity of their import markets, measured as the import-weighted firm average over 2004–2007 of the origin-sector(HS4)-year leave-one-out share of competitors' imports invoiced in US dollars, as in Equation (14), split into terciles. The right panel restricts the sample to exporting firms and groups them by natural hedging, measured as the firm's share of 2004–2007 export value invoiced in US dollars, distinguishing partially or un-hedged exporters (NH $<$ 1) from fully hedged exporters (NH $=$ 1). Data are from Garcia-Marin et al. (2019), 2004–2015.

A.11 Additional Results: Robustness & Financial Hedging

Table 17: Robustness: CIP Deviations

	OLS				IV	
	(1)	(2)	(3)	(4)	(5)	(6)
	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$	$Peso_{fopt}$
$CIP_t^{\$,PESO}$	0.121***	–	0.190***	–	0.417**	0.364**
	(0.042)	(.)	(0.041)	(.)	(0.197)	(0.182)
$SC_{kt}^{\$}$	-0.004***	-0.001	-0.001**	0.000	-0.018	-0.006
	(0.001)	(0.001)	(0.001)	(0.001)	(0.013)	(0.008)
$NH_{ft}^{\$}$	-0.012***	-0.011***	-0.012***	-0.011***	-0.012***	-0.012***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$SC_{kt}^{\$} \times CIP_t^{\$,PESO}$	-0.144***	-0.135***	-0.189***	-0.176***	-0.521**	-0.414*
	(0.023)	(0.023)	(0.021)	(0.021)	(0.246)	(0.229)
$NH_{ft}^{\$} \times CIP_t^{\$,PESO}$	0.066*	0.050	0.043	0.024	0.104**	0.067
	(0.037)	(0.037)	(0.038)	(0.038)	(0.046)	(0.044)
Firm-Origin-HS8	Yes	Yes	No	No	Yes	No
Firm-Origin	No	No	Yes	Yes	No	Yes
Firm-HS8	No	No	Yes	Yes	No	Yes
Year	No	Yes	No	Yes	No	No
Weak IV F-Stat					317.95	451.75
Observations	1054967	1054967	1284300	1284300	1054967	1284300

Note: The table reports the coefficients from the specification in Equation (13), replacing UIP deviations with CIP deviations in all columns. The dependent variable is a dummy variable equal to 1 when the $fopt$ transaction is invoiced in Chilean peso and 0 in U.S. dollar. $UIP_t^{\$,PESO}$ denotes deviations from UIP as defined in Equation (12). $CIP_t^{\$,PESO}$ denotes deviations from CIP computed using one-year-ahead forward exchange rates between the U.S. dollar and the peso. $SC_{kt}^{\$}$ is the strategic complementarity index, instrumented as in Equation (15) in Columns (5) and (6), and $NH_{ft}^{\$}$ is the share of firm level exports invoiced in U.S. dollars. Fixed effects are as reported in the Table: firm-product-origin in Columns (1), (2) and (5), and firm-origin together with firm-product in Columns (3), (4) and (6). Columns (2) and (4) include year fixed effects, which absorb the main effect of $CIP_t^{\$,PESO}$. Standard errors are clustered at the firm-product-origin level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

A.12 Additional Results: Euro Invoicing

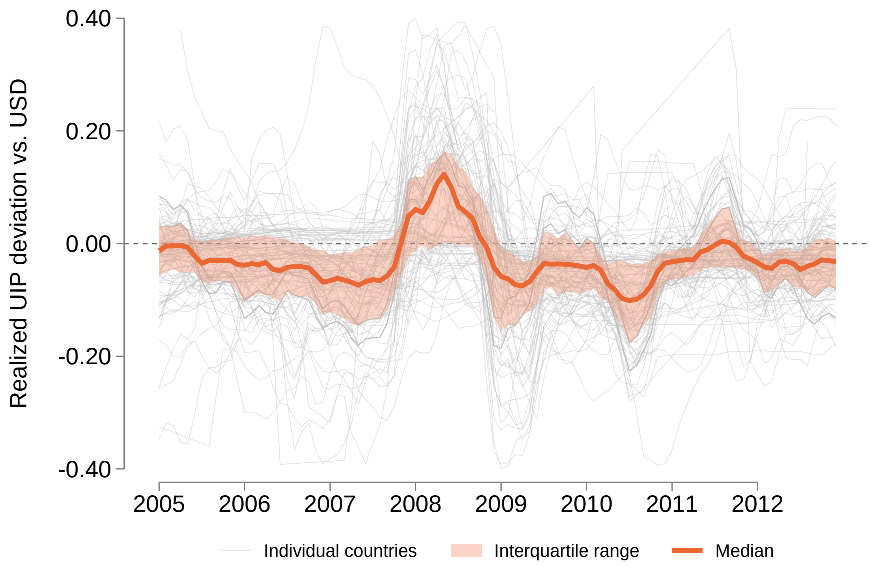
Table 18: The Euro Mirror of the Main Specification

	$Peso^{e}_{fopt}$			$Switch^{e}_{fopt}$		$Adopt^{e}_{fopt}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	OLS	IV	OLS	IV	OLS	IV
$UIP^{EUR,PESO}_{t}$	0.0337***	0.0430***	0.0708***	0.0835***	0.1379***	0.0533***	0.1035***
	(0.0020)	(0.0056)	(0.0214)	(0.0098)	(0.0407)	(0.0059)	(0.0229)
SC^{EUR}_{kt}		-0.0031***	-0.0094	-0.0062***	-0.0248*	-0.0030***	-0.0078
		(0.0009)	(0.0095)	(0.0017)	(0.0128)	(0.0009)	(0.0082)
NH^{EUR}_{fot}		-0.0124***	-0.0123***	-0.0249***	-0.0247***	-0.0135***	-0.0133***
		(0.0008)	(0.0008)	(0.0015)	(0.0015)	(0.0009)	(0.0009)
$SC^{EUR}_{kt} \times UIP^{EUR,PESO}_{t}$		-0.0707***	-0.1149***	-0.1321***	-0.2144***	-0.0856***	-0.1644***
		(0.0079)	(0.0339)	(0.0141)	(0.0620)	(0.0084)	(0.0359)
$NH^{EUR}_{fot} \times UIP^{EUR,PESO}_{t}$		-0.0941***	-0.0916***	-0.2332***	-0.2310***	-0.1237***	-0.1202***
		(0.0082)	(0.0085)	(0.0150)	(0.0152)	(0.0088)	(0.0090)
Firm-Origin-HS8 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weak IV F-stat			190.1		103.3		181.1
Observations	321,669	315,281	315,281	155,627	155,627	306,920	306,920

Notes: The sample is firm $\times$ origin $\times$ HS8 (fop) observations invoiced in euros or pesos over the period 2004–2013. The dependent variables are: an indicator for peso rather than euro invoicing (Columns 1–3); switching to the peso from euro without peso in the previous year (Columns 4–5); and first peso adoption on the censored never-adopted panel (Columns 6–7). $UIP^{EUR,PESO}_{t}$ is the realized 12-month euro–peso UIP deviation, constructed from the 3-month Euribor and the Chilean money-market rate with the realized depreciation of the peso against the euro; positive values mean euro funding is more expensive relative to peso funding. SC^{EUR}_{kt} is the leave-one-out share of competitors' imports invoiced in euros in the origin–HS4 market. NH^{EUR}_{fot} is the share of export invoiced in euros at the firm-origin level, set to zero when the firm has no exports to that origin. All regressions except Column (1) include an indicator for any exports to the origin and its interaction with $UIP^{EUR,PESO}_{t}$, so the natural hedge coefficient is identified off the euro-invoiced share among firms exporting to the origin. In all columns, we control for firm-level size (firm's share of aggregate Chilean imports). In the IV columns, SC^{EUR} and its interaction are instrumented by the just-identified leave-out pair built from competitors' euro export hedges. Standard errors, in parentheses, are clustered at the fop level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

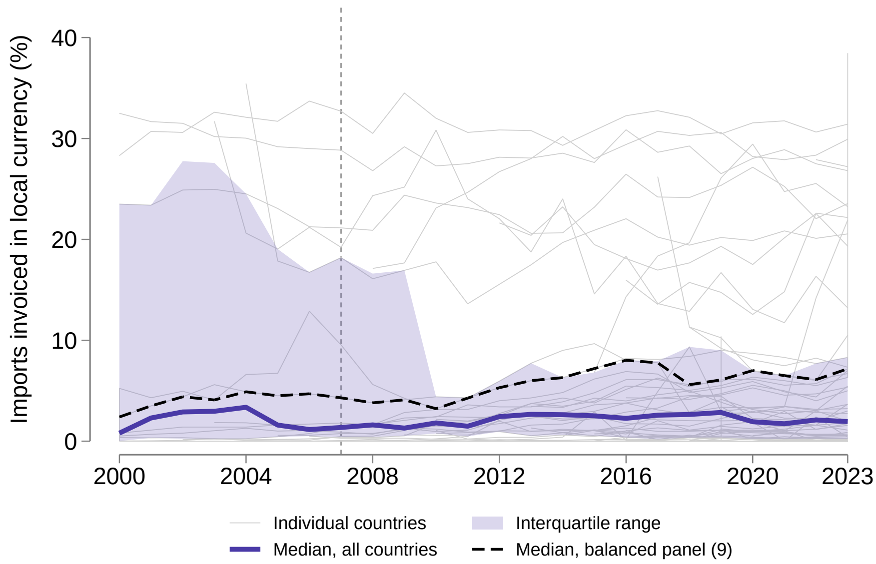
A.13 Additional Results: Cross-Country Evidence of Invoicing Currency

Figure 21: Realized UIP Deviations Across Countries



Note: The Figure plots the one-year realized UIP deviation against the US dollar for 113 countries, at monthly frequency, from January 2005 to December 2012, constructed as in Equation (12) using deposit rates from IMF International Financial Statistics and the three-month US certificate of deposit rate from FRED. Positive values indicate that borrowing in local currency is relatively cheaper than borrowing in US dollars. Data are monthly and smoothed using a two-month rolling window, as in Figure 5. The solid line is the cross-country median, the shaded area the interquartile range, and grey lines individual countries. The sample excludes currency unions and aggregates, euro-denominated economies, and long-standing hard pegs.

Figure 22: Local-Currency Invoicing of Imports Across Countries



Note: The Figure plots the share of imports invoiced in the home currency, in percent, for 62 countries between 2000 and 2023, from Boz et al. (2025). The sample excludes currency unions and aggregates, euro-denominated economies, and long-standing hard pegs; the database sets the home-currency share to missing wherever the home currency is the euro or the US dollar. The solid line is the median across all countries in each year, the shaded area the interquartile range, and grey lines individual countries. The panel expands from 10 countries in 2000 to 58 in 2023, and the economies entering after 2010 invoice a small share of imports at home, so the unconditional median and interquartile range are not comparable over time: the dashed line reports the median across the nine economies observed in every year, which holds composition fixed and rises from about 3 percent around 2010 to about 8 percent by the mid-2010s. The vertical dashed line marks 2007. Eleven observations exceed 40 percent and are omitted from the figure but used when computing the median and the interquartile range.

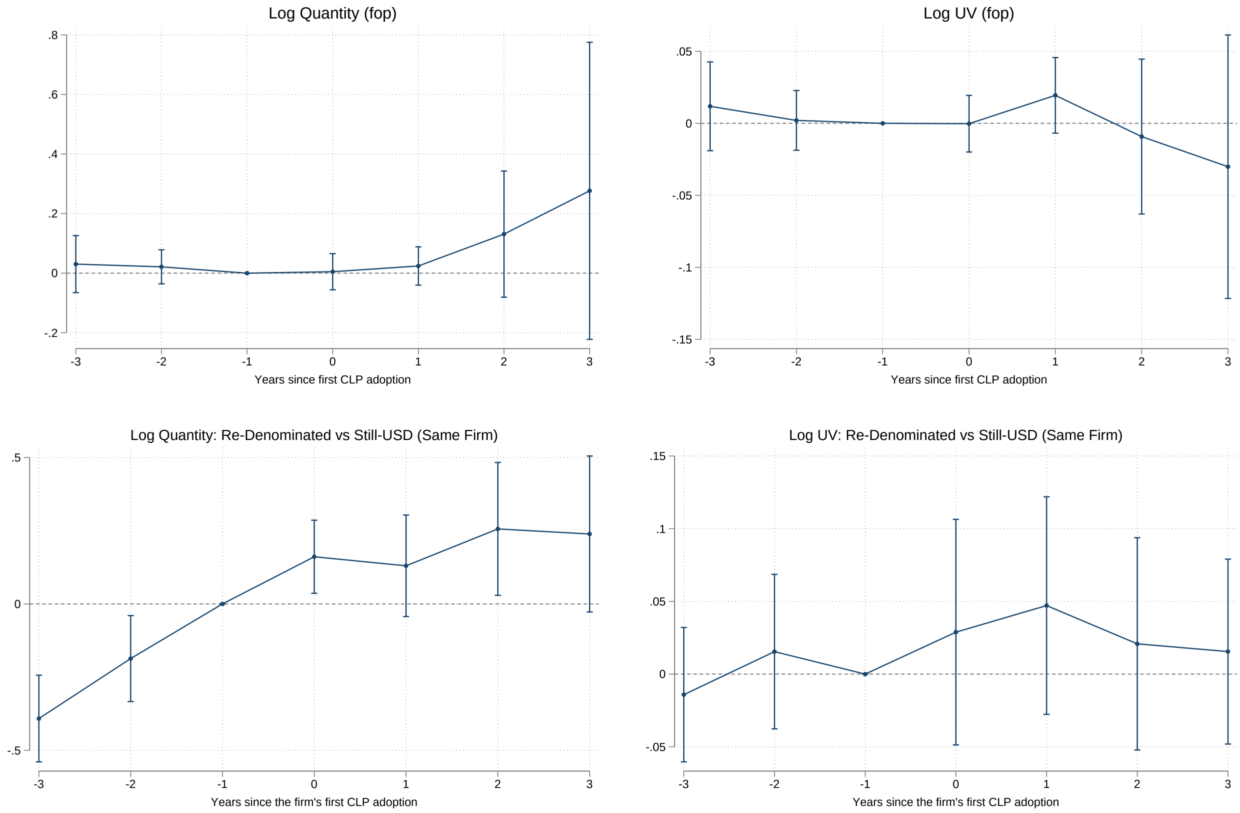
Table 19: Local-currency Import Invoicing and Realized UIP Deviation

	(1) OLS	(2) Country FE	(3) Two-way FE	(4) EM
UIP deviation vs. USD	0.00678	0.549	1.800	1.962
	(3.409)	(1.474)	(1.246)	(1.447)
Observations	555	552	552	425
Country FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Sample	All	All	All	EM
Countries	56	53	53	45

Note: The Table reports the country-level counterpart of Equation (13). The dependent variable is the share of imports invoiced in the home currency, in percent, from Boz et al. (2025). $UIP_{ct}^{\$,LC}$ is the realized UIP deviation of country c against the US dollar, constructed as in Equation (12) and averaged to annual frequency. Column (1) is pooled OLS; Column (2) adds country fixed effects; Column (3) adds year fixed effects; Column (4) estimates the specification of Column (3) on emerging markets only, classified according to the IMF World Economic Outlook. The sample covers 2000–2021, the period over which both series are available, and excludes currency unions and aggregates, the United States, euro-denominated economies and long-standing hard pegs. Standard errors are clustered at the country level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.14 Additional Results: Real Effects of Peso Adoption

Figure 23: Quantities and Prices around First Peso Adoption



Note: The Figure plots event-study coefficients for log quantity (left column) and log unit value (right column) at the firm-origin-product(HS8) combination (fop) level, by year relative to the firm's first year of Chilean peso invoicing. Adopters are defined as firms f with first peso use in 2008–2012 that previously imported exclusively in US dollars. The top panels ask whether an adopter's fop combinations change when the firm starts invoicing in pesos: each point compares the adopter's fop combinations with those of firms that never invoice in pesos, within the same origin-sector(HS4)-year market, τ years from adoption and relative to the year before adoption. These estimates average over all of the adopter's fop combinations, whether or not their own invoicing changes. The bottom panels look inside the adopting firm and ask whether the fop combinations that change currency behave differently from those that keep invoicing in dollars: each point is the gap between the re-denominated and the still dollar-invoiced fop combinations of the same firm f in the same year, relative to that gap in the year before adoption. All specifications include fop fixed effects. The bottom panels use only the adopter's fop combinations observed before the adoption year. Event time is binned at the ± 3 endpoints and $\tau = -1$ is the omitted category in all panels. All coefficients are in log points. Data are from Garcia-Marin et al. (2019), 2004–2015. We report 95 percent confidence intervals. Standard errors are clustered at the firm level.

B Appendix: Theory

B.1 Firm Invoicing Choice.

Start from the firm maximization problem displayed in Equation (9). For any given η_i, the optimal domestic price $p_i^{d\star} = \frac{\theta}{\theta-1}E[C_i(s^v, \eta_i)]$. A firm invoices working capital x_i in vehicle currency v ($\eta_i^\star = 0$) if and only if expected profit for doing so is higher or equal to the expected profit in the complementary case when working capital is invoiced in domestic currency (($\eta_i^\star = 1$)). That is:

$$
\begin{aligned}
&\eta_i^\star = 0 \qquad \text{iff} \\
&p_i^d \left(p_i^d\right)^{-\theta} - \mathbb{E}\left[C_i(s^v, 0)\right] \left(p_i^d\right)^{-\theta} - F_i^v \geq p_i^d \left(p_i^d\right)^{-\theta} - \mathbb{E}\left[C_i(s^v, 1)\right] \left(p_i^d\right)^{-\theta} - F_i^d
\end{aligned}
$$

Plug in the optimal price expression for the respective cases and obtain:

$$
\begin{aligned}
&\frac{\theta}{\theta-1}\mathbb{E}[C_i(s^v,0)] \left(\frac{\theta}{\theta-1}\mathbb{E}[C_i(s^v,0)]\right)^{-\theta} - \mathbb{E}\left[C_i(s^v,0)\right] \left(\frac{\theta}{\theta-1}E[C_i(s^v,0)]\right)^{-\theta} - F_i^v \geq \\
&\frac{\theta}{\theta-1}\mathbb{E}[C_i(s^v,1)] \left(\frac{\theta}{\theta-1}\mathbb{E}[C_i(s^v,1)]\right)^{-\theta} - \mathbb{E}\left[C_i(s^v,1)\right] \left(\frac{\theta}{\theta-1}\mathbb{E}[C_i(s^v,1)]\right)^{-\theta} - F_i^d
\end{aligned}
$$

Which simplifies to:

$$
\frac{(\theta-1)^{\theta-1}}{\theta^\theta}\left[\mathbb{E}[C_i(s^v,0)]^{1-\theta} - \mathbb{E}[C_i(s^v,1)]^{1-\theta}\right] \geq F_i^v - F_i^d
$$

Then, consider the expression for the expected marginal cost of the firm for the two different cases:

$$
\begin{aligned}
\mathbb{E}[C_i(s^v,0)] &= w^{1-\alpha_i}\mathbb{E}\left[((1+i_v)\rho_v s_v)^{\alpha_i}\right] \\
\mathbb{E}[C_i(s^v,1)] &= w^{1-\alpha_i}\left[(1+i_d)\rho_d\right]^{\alpha_i}
\end{aligned}
$$

Re-write the expression above as:

$$
\begin{aligned}
&\mathbb{E}\left[((1+i_v)\rho_v s_v)^{\alpha_i}\right]^{1-\theta} - \left[(1+i_d)\rho_d\right]^{\alpha_i(1-\theta)} \geq \tau_i \times \left(F_i^v - F_i^d\right) \\
&\tau_i \equiv \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{1}{w^{(1-\alpha_i)(1-\theta)}}
\end{aligned}
$$

which is the expression in the main text. Note that $\frac{\theta^\theta}{(\theta-1)^{\theta-1}} > 1$ for $\theta > 1$ and it is strictly increasing in θ.

Then, consider the simplified case in which $F_i^v = F_i^d$ and $s^v \sim$ Log-Normal(μ, Σ). Then,

$$
((1+i_v)\rho_v)^{\alpha_i(1-\theta)}\,\mathbb{E}\left[s_v^{\alpha_i}\right]^{1-\theta} \geq \left[(1+i_d)\rho_d\right]^{\alpha_i(1-\theta)}
$$

Then, i) elevate both sides to the power of $1/(1-\theta)$, ii) take logs on both sides, iii) use the result that $\mathbb{E}\left[s_v^{\alpha_i}\right] = \exp(\alpha_i \mu + 0.5 \times \alpha_i^2 \Sigma)$, and obtain the expression in the text.

B.2 Proof of Proposition 1.

Manipulating Equation (10) we can write:

$$A(1+i_v)^{\alpha_i(1-\theta)} - B > C \qquad \text{with } A \equiv \mathbb{E}\left[(\rho_v s_v)^{\alpha_i}\right]^{1-\theta}; B \equiv \left[\rho_d(1+i_d)\right]^{\alpha_i(1-\theta)}; C \equiv \tau_i\left(F_i^v - F_i^d\right). \tag{25}$$

Changing i_v scales only the first term in the left-hand side. Given that $\alpha_i > 0$ and $\theta > 1$, increasing i_v lowers the left-hand side and therefore reduces the chance the inequality holds and the probability of invoicing in vehicle currency.

B.3 Proof of Proposition 2.

Further manipulating Equation (25), we obtain the threshold i_v^* that makes importers indifferent between vehicle and local currency invoicing:

$$i_v^* = \exp\left(\frac{\log(B+C)}{\alpha_i(1-\theta} - \log\rho_v - \mu - 0.5\alpha_i\Sigma\right) - 1, \tag{26}$$

where we used the fact that s_v is distributed according to a log-normal with mean μ and variance Σ. We prove Proposition 2 by showing that the threshold i_v^* is decreasing in α_i. Increasing i_v makes the condition $\eta_i^* = 0$, i.e. vehicle currency invoicing, less likely. The threshold determines the cutoff, so how α_i affects the sensitivity of the 0-1 outcome to changes in i_v is captured by how the threshold i_v^* moves with α_i. It can be shown that:

$$\frac{\partial i_v^*}{\partial \alpha} \equiv (1+i_v^*)\left(\frac{\log(B+C)}{\alpha_i^2(\theta-1)} - 0.5\Sigma\right).$$

Proposition 2 predicts that $\frac{\partial i_v^*}{\partial \alpha} < 0$, i.e. higher α_i lowers the threshold, reducing the region in which $\eta_i^* = 0$ and making the probability of vehicle currency lower when i_v rises. The sign of the derivative above is negative for sufficiently high variance of the exchange rate Σ. This is in line with the key intuition. Firms maintain a constant markup over marginal costs, so vehicle-currency invoicing implies that ex-post marginal costs depend on the realized exchange rate, potentially generating costly deviations from the optimal markup. Thus, firms choose vehicle-currency invoicing only when its financing cost advantage outweighs the risk of markup misalignment, which ultimately depends on the volatility Σ.

B.4 Proof of Proposition 3.

We can prove Proposition 3 by applying the same reasoning as above and differentiating Equation (26) with respect to the fixed cost F_v:

$$\frac{\partial i_v^*}{\partial F_v} \equiv \frac{\tau_i}{\alpha_i(1-\theta)} \frac{(1+i_v^*)}{B+C} < 0.$$

It follows that lower F_v, or equivalently stronger strategic complementarities in vehicle currency invoicing, loosen the inequality making $\eta_i^* = 0$ more likely.

C A Model of Currency Choice: Exporter Extension

In this section, we present an alternative version of the theoretical framework in which firms export too rather than only import. Every element of the model is identical to Section 4 except for a single change in assumption: firms sell exclusively to one foreign destination market, and all revenues are denominated in US dollars (USD, the vehicle currency). We show the implications of this single change through the entire model, deriving the analogue of each result from the baseline.

C.1 Environment

A small open economy is populated by a continuum of exporting firms indexed by $i \in [0, 1]$. Each firm produces a differentiated good sold exclusively in a single foreign destination market and sets prices in vehicle currency (USD). Demand for firm i's output in the foreign market is given by

$$y_i^f = \left(\frac{p_i^v}{q^f}\right)^{-\theta},$$

where p_i^v is the firm's USD price, q^f is a destination-market demand shifter (normalised to one below), and $\theta > 1$ is the constant elasticity of demand.

Production combines domestic labor, l_i, and imported intermediate inputs, x_i, according to the same constant-returns-to-scale Cobb–Douglas technology as in the baseline:

$$y_i = A_i\, l_i^{1-\alpha_i} x_i^{\alpha_i}, \tag{27}$$

where $\alpha_i \in (0, 1)$ is the firm-specific share of imported inputs in production, and $A_i \equiv (\alpha_i^{-1})^{\alpha_i}((1-\alpha_i)^{-1})^{1-\alpha_i}$ is a normalisation constant. Imported inputs are working capital and must be financed prior to production while labor costs are settled after revenues are realised.

Time is divided into a morning and an evening within each period, as in the baseline. In the morning, firms choose USD export prices which are nominally rigid and finance imported inputs. In the evening, the bilateral exchange rate s_v (domestic currency per vehicle currency) is realised. Then, firms purchase inputs, produce, export at the pre-set price, collect revenues in USD, and repay their loans.

Revenue denomination. Because the firm invoices its exports in USD, revenues expressed in domestic currency are stochastic: $\text{Revenue}_i = s_v\, p_i^v\, (p_i^v)^{-\theta}$. This is the key difference from the baseline, in which revenues in domestic currency are deterministic. The exchange rate realisation therefore generates fluctuations on the revenue side for the exporter.

The imported-input technology, the financing options, the ex-post marginal cost of production, and the fixed-cost structure are identical to those in Section 4:

C.2 Working Capital Invoicing Choice

Optimal Invoicing Decision. Firms maximise expected profits in domestic currency. Because export prices are set in USD before s_v is realised, expected profits are

$$\max_{\eta_i, p_i^v} \mathbb{E}[\pi_i^j] = \mathbb{E}[s_v]\,(p_i^v)^{1-\theta} - \mathbb{E}[C_i(s_v,\eta_i)]\,(p_i^v)^{-\theta} - \eta_i F_i^d - (1-\eta_i)F_i^v. \tag{28}$$

For a given η_i, the optimal USD export price is

$$p_i^{v\star} = \frac{\theta}{\theta-1}\,\frac{\mathbb{E}[C_i(s_v,\eta_i)]}{\mathbb{E}[s_v]}, \tag{29}$$

which is the standard constant-markup rule scaled by the expected domestic-currency value of one dollar. Substituting (29) into (28), expected profits equal

$$\mathbb{E}[\pi_i^j] = \frac{(\theta-1)^{\theta-1}}{\theta^\theta}\,\mathbb{E}[C_i(s_v,\eta_i)]^{1-\theta}\,\mathbb{E}[s_v]^\theta - F_i^j. \tag{30}$$

Compared with the baseline (Equation (9)), the $\mathbb{E}[s_v]^\theta$ factor multiplies the profit term. Because this factor is common to both currency options, it cancels in the relative comparison. The firm chooses to invoice imported inputs in vehicle currency ($\eta_i^\star = 0$) if and only if

$$\mathbb{E}[(\rho_v(1+i_v)s_v)^{\alpha_i}]^{1-\theta} - [\rho_d(1+i_d)]^{\alpha_i(1-\theta)} > \tilde{\tau}_i\left(F_i^v - F_i^d\right), \tag{31}$$

where

$$\tilde{\tau}_i \equiv \frac{\tau_i}{\mathbb{E}[s_v]^\theta} = \frac{\theta^\theta}{(\theta-1)^{\theta-1}} \times \frac{w^{(1-\alpha_i)(1-\theta)}}{\mathbb{E}[s_v]^\theta}, \tag{32}$$

and τ_i is the adjustment factor from the baseline (Equation (10)). See Appendix C.3 for the derivation.

Comparison with the baseline. The left-hand side of (31) is identical to that in the baseline condition in Equation (10). The only difference is that the effective fixed-cost weight on the right-hand side, $\tilde{\tau}_i$, is scaled down by $\mathbb{E}[s_v]^\theta$. Intuitively, exporters with USD revenues benefit from USD-denominated costs because revenues and costs co-move with the exchange rate, providing a natural hedge. This natural-hedge motive reduces the effective cost of vehicle-currency invoicing without altering the direction of any comparative-static result.

UIP-deviation reformulation. Setting $F_i^v = F_i^d$ and assuming $s_v \sim$ Log-Normal(μ, Σ), the cutoff condition in Equation (31) reduces to

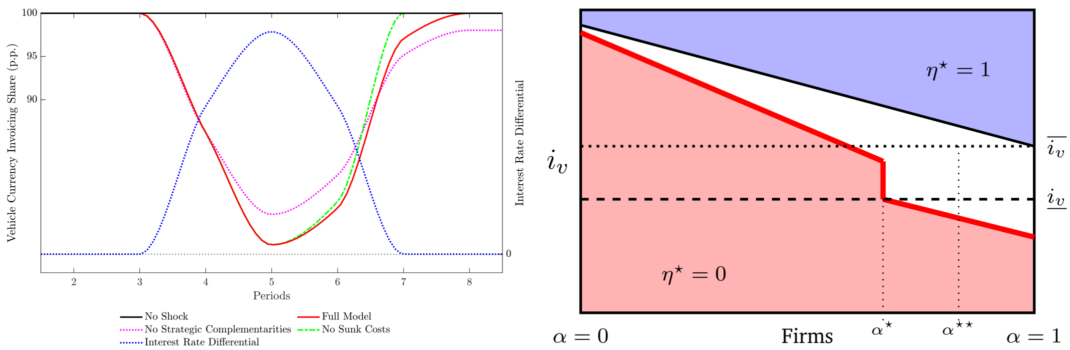
$$\eta_i^\star = 0 \qquad \text{iff} \qquad \underbrace{i_v - i_d + [\mu + 0.5\,\alpha_i \Sigma]}_{\text{UIP Deviation}} < \log\left(\frac{\rho_d}{\rho_v}\right), \tag{33}$$

which is identical in form to Equation (11) in the baseline. When fixed costs differ across currencies, the advantage of vehicle-currency invoicing is amplified by the $1/\mathbb{E}[s_v]^\theta$ scaling in $\tilde{\tau}_i$, making vehicle-currency invoicing strictly more attractive than in the baseline for any given level of i_v.

The three propositions from the baseline carry over unchanged in their mathematical statements. Only the economic intuition for Proposition 2 is different. Indeed, the mathematical proof of Proposition 2 is identical to that in Appendix B (the cutoff LHS is unchanged). The economic mechanism, however, differs from the baseline. In the baseline, revenues in domestic currency create a natural-hedge motive against vehicle-currency costs; high-α_i firms are most exposed to exchange-rate risk and therefore most sensitive to i_v. In the exporter extension, revenues are in vehicle currency, so vehicle-currency costs actually hedge the revenue risk. Nevertheless, the financing-cost channel is still active: higher α_i implies a larger share of production costs that respond to i_v. When i_v rises, higher-α_i exporters face a larger increase in expected marginal costs, making domestic-currency invoicing more attractive. The α_i-amplification of Proposition 2 therefore survives.

Lastly, the proof and mechanism of Proposition 3 are identical to the baseline: lower F_i^v (from stronger complementarities or lower sunk costs) reduces the right-hand side of (31), making vehicle-currency invoicing more attractive. The complementarities are amplified in the exporter extension because the baseline effective threshold is already lower: any given reduction in F_i^v therefore has a proportionally larger effect on the invoicing decision.

Figure 24: Shock to Vehicle Currency Financing (Exporter Extension)



Note: The right panel plots the optimal invoicing decision for each level of i_v (y-axis) conditional on α (x-axis). The left panel shows the dynamics of the aggregate USD invoicing share. The red line is the full model; the green line abstracts from sunk costs; the purple line abstracts from strategic complementarities; the blue line shows the interest rate differential $i_v - i_d$. Calibration: $\rho_v = \rho_d = 1$; $\theta = 5$; $w = 0.2$; $\alpha_i \sim U[0, 1]$; $\gamma^v = 0.7$; $\gamma^d = 0$; $\gamma_1 = 2.5$; $\kappa_0^v = 1$; $f^d = \{0.2, 0\}$; initial differential $\underline{i_v} = 0.55$; peak shock $\overline{i_v} = 0.67$.

C.3 Temporary Shocks, Invoicing Dynamics and Hysteresis

We now examine the dynamic response to a temporary rise in i_v in the exporter model. The mechanism is qualitatively identical to the baseline (Section 4.3): temporary shocks generate permanent invoicing changes through sunk costs and strategic complementarities. The quantitative magnitude differs: because vehicle-currency invoicing is more attractive for exporters at the initial steady state (lower effective threshold), a larger shock is required to induce switching, and the post-shock share of vehicle-currency invoicing remains higher.

We use the same calibration as in Section 4.3 and assume that all firms initially invoice in vehicle currency. Figure 24 shows that, in line with the empirical patterns, the same UIP shocks that moves invoicing patterns on the importer side does not have any long-lasting effect on the exporter side.

Short-run response. The right panel of Figure 24 shows the firm-level invoicing decision. For a given i_v, firms with $\alpha_i > \tilde{\alpha}$ invoice in vehicle currency ($\eta^\star = 0$) and those with $\alpha_i < \tilde{\alpha}$ invoice in domestic currency ($\eta^\star = 1$). The downward-sloping boundary reflects Proposition 2: higher-α_i exporters are more sensitive to i_v because a greater share of their production costs is exposed to the financing-cost channel.

Relative to the baseline (Figure 4), the vehicle-currency region ($\eta^\star = 0$) is larger at every i_v, consistent with the lower effective fixed-cost threshold $\tilde{\tau}_i < \tau_i$. The shock from $\underline{i_v}$ to $\overline{i_v}$ causes firms with $\alpha_i > \alpha^{\star\star}$ to switch to domestic-currency invoicing. However, the mass of switchers is smaller than in the baseline for an equally sized shock, reflecting the stronger attachment of exporters to vehicle-currency invoicing.

Hysteresis in invoicing. The sunk-cost mechanism operates exactly as in Section 4.3: once a firm switches to domestic-currency invoicing, the sunk cost permanently lowers the fixed cost of domestic invoicing, preventing reversion when i_v returns to $\underline{i_v}$. The red line in the left panel of Figure 24 shows that the aggregate vehicle-currency invoicing share does not return to one after the shock, producing the same hysteresis documented in the baseline. For the same reason as in Section 4.3, strategic complementarities amplify the initial decline (more firms piggyback on early switchers), while sunk costs prevent reversion.

The key quantitative difference is that the initial steady-state share is fully at one (all exporters invoice in USD) and the post-shock share is closer to one than in the analogous baseline experiment. This reflects the natural-hedge benefit captured by $\tilde{\tau}_i$: exporters require a larger or more persistent shock to shift away from vehicle-currency invoicing permanently.

Appendix: Derivation of Extension Propositions

Derivation of Equation (31). Starting from the firm's maximisation problem in Equation (28), the optimal USD price for a given η_i is $p_i^{v\star} = \frac{\theta}{\theta-1}\frac{\mathbb{E}[C_i(s_v,\eta_i)]}{\mathbb{E}[s_v]}$. Substituting into expected profits:

$$\begin{aligned}\mathbb{E}[\pi_i^j] &= \mathbb{E}[s_v]\,(p_i^{v\star})^{1-\theta} - \mathbb{E}[C_i]\,(p_i^{v\star})^{-\theta} - F_i^j \\ &= \frac{(\theta-1)^{\theta-1}}{\theta^\theta}\,\mathbb{E}[C_i(s_v,\eta_i)]^{1-\theta}\,\mathbb{E}[s_v]^\theta - F_i^j.\end{aligned}$$

Firm i chooses $\eta_i^\star = 0$ iff $\mathbb{E}[\pi_i^v] \geq \mathbb{E}[\pi_i^d]$, i.e.

$$\mathbb{E}[C_i(s_v,0)]^{1-\theta} - \mathbb{E}[C_i(s_v,1)]^{1-\theta} \;\geq\; \frac{\theta^\theta}{(\theta-1)^{\theta-1}}\frac{F_i^v - F_i^d}{\mathbb{E}[s_v]^\theta}.$$

Substituting $\mathbb{E}[C_i(s_v,0)] = w^{1-\alpha_i}\mathbb{E}[(\rho_v(1+i_v)s_v)^{\alpha_i}]$ and $\mathbb{E}[C_i(s_v,1)] = w^{1-\alpha_i}[\rho_d(1+i_d)]^{\alpha_i}$, and dividing by $w^{(1-\alpha_i)(1-\theta)}$, yields (31)–(32). The proofs of Propositions 1–3 follow directly from those in Appendix B, since the LHS of (31) is identical to that of the baseline condition (10).